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Contextual Stochastic Optimization for Omnichannel Multicourier Order Fulfillment Under Delivery Time Uncertainty

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Abstract. ***Problem definition*:** The paper studies a large-scale order fulfillment problem for a leading e-commerce company in the United States. The challenge involves selecting fulfillment centers and shipping carriers with observational data only to efficiently process orders from a vast network of physical stores and warehouses. The company's current practice relies on heuristic rules that choose the cheapest fulfillment and shipping options for each unit without considering opportunities for batching items or the reliability of carriers in meeting expected delivery dates. ***Methodology/results*:** The paper develops a data-driven contextual stochastic optimization (CSO) framework that integrates distributional forecasts of delivery time deviations with stochastic and robust order fulfillment optimization models. The framework optimizes the selection of fulfillment centers and carriers, accounting for item consolidation and delivery time uncertainty. Validated on a real-world data set containing tens of thousands of products, each with hundreds to thousands of fulfillment options, the proposed CSO framework significantly enhances the accuracy of meeting customer-expected delivery dates compared with current practices. It provides a flexible balance between reducing fulfillment costs and managing delivery time deviation risks, emphasizing the importance of contextual information and distributional forecasts in order fulfillment. ***Managerial implications*:** This is the first study of an omnichannel multicourier order fulfillment problem with delivery time uncertainty through the lens of contextual optimization, fusing machine learning and optimization. The results offer actionable guidance for retailers to enhance service quality and customer satisfaction while balancing cost efficiency and risk, supporting higher retention and profitability.

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1. Introduction

Over the past decade, online shopping has surged dramatically. The e-commerce share of total retail sales in the United States has risen from 6% in 2015 to 16% in 2024, exceeding 1.1 trillion dollars (U.S. Department of Commerce 2024). Amid intense competition in pricing, service, and marketing driven by the lucrative e-commerce market, logistical performance has become a key element to success. Last-mile fulfillment, in particular, is critical as it accounts for 55% of the total transportation costs in the fulfillment process (Viswanathan 2023). This evolving market landscape poses unprecedented challenges for e-commerce companies striving to deliver satisfactory customer services. Fortunately, the growing volume of data also offers opportunities for data-driven operational improvements, enabling companies to boost revenue and cut costs (Fisher et al. 2019).

In addition, after the coronavirus disease 2019 pandemic, the demand for timely home delivery has also increased significantly. As a result, omnichannel fulfillment and the use of multiple couriers, including crowd shipping, have become essential components of the fulfillment landscape, drawing significant attention both in industry and in academia (Dethlefs et al. 2022, Das et al. 2023, Mohri et al. 2023). The omnichannel approach combines distribution centers (DCs) and brick-and-mortar stores to fulfill online orders. Unlike the traditional method that uses DCs for online orders and stores for in-store purchases separately, the omnichannel approach leverages the proximity of physical stores to delivery addresses. This can potentially offer

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more cost-effective and faster delivery options for online customers (Acimovic and Farias 2019). Meanwhile, in the crowd-shipping model, retailers recruit nondedicated drivers, such as gig couriers like Roadie, to handle pickup and delivery tasks for online customers. This approach can alleviate the workload of employed drivers during peak hours and offer faster delivery to customers, resulting in higher service satisfaction (Zehtabian et al. 2022, Behrendt et al. 2023). *The problem considered in this paper involves fulfilling orders by choosing from various types of fulfillment centers, such as warehouses and stores, and different couriers, including traditional shipping companies and gig couriers.*

Given that transportation and shipping are significant cost drivers in e-commerce (Hübner et al. 2013, Kuhn and Sternbeck 2013, Dethlefs et al. 2022), delivery consolidation has emerged as an effective strategy for fulfilling orders with multiple items. This approach has been shown to enhance consumer satisfaction (Wagner et al. 2023) while also reducing shipping costs for online retailers (Chen et al. 2024). In addition to cutting parcel expenses, item consolidation enhances the customer experience by eliminating the need to receive multiple packages for a single order. This approach not only improves convenience but also promotes a more sustainable and environmentally friendly fulfillment process by decreasing the number of packages shipped and the number of delivery trips needed. This reduction in shipments and trips leads to lower carbon emissions and less packaging waste, contributing to greener logistics practices (Ülkü 2012). *This paper also investigates multi-item order consolidation in its omnichannel fulfillment problem.*

Most existing studies on order fulfillment optimization have not explicitly considered the uncertainty in delivery timeliness. However, Freedman (2019) highlights the strong desire of customers for accurate delivery time promises and faster service. The survey indicates that around 44% of online shoppers have abandoned their shopping carts because the items that they wanted would not arrive on time, whereas 20% abandon orders with unclear delivery dates. Furthermore, 28% of shoppers are willing to pay for expensive, expedited shipping to receive products at their desired time. The statistics clearly show that customers prioritize service quality, sometimes even more than the product value, when making purchase decisions. Moreover, Salari et al. (2022) point out the importance of considering delivery time uncertainty in order fulfillment to better meet customer expectations and improve satisfaction. Their proposed promised delivery time policy demonstrates a 6.1% increase in sales volume in numerical experiments. Notably, overpromising delivery times can lead to higher return rates, lower repurchase rates, and dissatisfaction with delayed service. Additionally, customers can have mixed sentiments over underpromised delivery times as early deliveries may not align with their availability. Therefore, missing the exact promised delivery date can negatively impact short-term and long-term sales, both implicitly and explicitly (Salari et al. 2022, Cui et al. 2024). *Given the crucial role of delivery timeliness, the paper incorporates delivery time uncertainty into the omnichannel multicourier order fulfillment optimization problem through a contextual stochastic optimization (CSO) framework.*

One of the key challenges of omnichannel multicourier order fulfillment optimization comes from the nature of historical data; only observational data (i.e., delivery time deviations for the fulfillment options actually chosen) are available. Counterfactual deviations for unchosen options are never observed, similar to the bandit feedback setting (Lattimore and Szepesvári 2020). *The availability of observational data only raises fundamental challenges that are typically not addressed in prior work.*

To address these challenges, *this paper proposes a contextual stochastic optimization framework, encompassing both risk-neutral and robust approaches.* Unlike traditional stochastic programming methods that depend on static, predefined probability distributions for uncertainties, CSO combines machine learning techniques with mathematical optimization to dynamically predict uncertainties based on contextual information. Thanks to the availability of contextual information, CSO has the flexibility of generating more tailored and effective decisions, leveraging the most current and relevant information. Such flexibility is particularly beneficial in the dynamic environment of online retail, where conditions and customer behaviors can change rapidly, requiring a more responsive approach to decision making. The effectiveness of CSO has been validated in the existing operations research and management science literature (Mišić and Perakis 2020, Sadana et al. 2025). This approach has proven successful in various applications, including inventory control (Bertsimas et al. 2016, Meller et al. 2018), price and revenue management (Ito and Fujimaki 2016, Perakis et al. 2023), and more recently, school redistricting (Guan et al. 2025).

1.1. Contributions

The key novelty in this paper is a CSO framework for an omnichannel multicourier order fulfillment optimization with observational data only. Existing studies on CSO typically assume the availability of a data set $\mathcal{D} = \{(\mathbf{s}_1, \mathbf{c}_1), (\mathbf{s}_2, \mathbf{c}_2), \ldots\}$, pairing covariates with fully observed uncertain parameter vectors. However, real-world scenarios often involve high-dimensional uncertain parameters where individual data points capture only partial information, particularly when uncertainty measurement depends on the chosen solution, which is the case in the application considered in this paper.

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To address the challenge of observational data, this paper introduces a generalized CSO framework addressing these real-world complexities. *The novel CSO framework is applied to a sophisticated stochastic omnichannel order fulfillment optimization problem with multiple carrier options and delivery time uncertainties*. In collaborating with a major U.S. online retailer, the framework is rigorously evaluated on an industrial order fulfillment data set featuring a vast, intricate fulfillment network. This application demonstrates the framework capabilities and its potential impact on large-scale, real-world optimization challenges. Based on the significant performance improvements demonstrated in this study, the industrial partner is actively considering a pilot program to integrate key elements of the proposed framework into their production fulfillment optimization engine. The contributions of the paper can be summarized as follows.

1. *A modular and scalable CSO framework for observational data*. The paper presents a novel, distribution-agnostic contextual stochastic optimization framework that converts problems with missing-counterfactual observational data into tractable problems. A contextual distribution oracle is learned from partial observations, and the resulting problem is solved via two scalable solution paradigms—contextual sample average approximation (C-SAA) and contextual robust optimization (C-RO). This features a tractable formulation for assignment-type problems with piecewise linear objectives.

2. *Tailored contextual distribution learning for e-commerce delivery time deviations*. The paper adapts calibrated probabilistic multiclass classification and tree-based quantile regression (QR) to exploit the discrete, ordinal nature of delivery time deviations, yielding well-calibrated contextual distributions for the CSO framework.

3. *Consolidation- and timeliness-aware mixed-integer linear program (MILP) for omnichannel multicourier fulfillment*. The paper proposes the first data-driven MILP that jointly selects fulfillment centers and carriers and captures item-consolidation discounts while embedding learned contextual deviation distributions in a tractable way.

4. *Industry-scale real-world validation*. The paper includes an extensive case study at a major e-commerce retailer to (1) validate the efficacy of the proposed approaches in reducing expected fulfillment costs and (2) show significant improvements in on-time delivery rates compared with standard heuristics and baseline models. These results demonstrate the potential for substantial real-world impact.

1.2. Structure of the Paper

The remainder of this paper is organized as follows. Section 2 reviews related literature. Section 3 formulates the omnichannel multicourier order fulfillment problem under delivery time deviation uncertainty. Section 4 develops a generic contextual stochastic optimization framework for observational data and introduces two tractable solution paradigms—C-SAA and C-RO. Section 5 describes the machine learning methods (multiclass classification and tree-based quantile regression) used to implement the contextual distribution oracle. Section 6 presents the case study, computational results, and actionable insights from a real data set. Finally, Section 7 concludes and outlines directions for future research.

2. Related Literature

There are several streams of literature related to this paper in terms of methodology and problem context.

2.1. Data-Driven Decision Making

This paper adds to the growing body of literature on data-driven decision making, particularly at the intersection of machine learning and optimization, which has led to the development of CSO. Sadana et al. (2025) put forth a comprehensive survey on contextual optimization. Tutorials on various methodologies and applications of CSO models in logistics and operations management are available in the works of Mišić and Perakis (2020), Qi and Shen (2022), and Tian et al. (2023). Among the body of research on CSO, the most relevant to this paper are those that focus on learning contextual distributions of uncertain parameters and incorporating these forecasts into optimization models. Data-driven approaches typically build these forecasts from empirical residual errors (Deng and Sen 2022, Perakis et al. 2023, Kannan et al. 2024) or weighted empirical distributions based on proximity to the training data (Bertsimas and Kallus 2020, Notz and Pibernik 2022). However, these studies assume full observation of uncertain parameters for each data point, which differs from the context of this paper. Recent research has also explored integrating downstream optimization effects into machine learning model training. Elmachtoub and Grigas (2022) introduce the smart predict then optimize (SPO) loss and its convex surrogate for point prediction models, with subsequent extensions to contextual distributional estimation (Qi et al. 2021, Kallus and Mao 2023). However, the SPO loss framework is not directly applicable to this paper's context as the data set analyzed here is observational. In such data sets, outcomes are only partially observed based on decisions made, presenting unique challenges not addressed by these existing methods.

This paper thus contributes to the field of data-driven decision making that handles observational data, a domain explored by prior research (e.g., in Bertsimas et al. 2019 and Jo et al. 2021) using optimal

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prescriptive trees for prescriptive analytics in personalized treatment. Although these methods effectively learn policies mapping features to actions, they cannot be directly applied to the fulfillment problem considered in this paper because of the presence of instance-specific hard constraints on decisions. The proposed framework addresses these limitations, offering a novel approach to CSO problems with observational data and constrained decision spaces.

Unlike earlier CSO applications that focus on risk-neutral objectives, a growing stream of research is concerned with risk-averse optimization models that incorporate contextual information (Bayram et al. 2022, Peršak and Anjos 2023, Patel et al. 2024, Sun et al. 2024). Building on these studies, this paper also adapts traditional robust optimization models to tackle a unique CSO problem that differs from those explored in other research.

2.2. Order Fulfillment Problem

The novelty in this paper is the application of CSO to a unique instance of the order fulfillment problem. The order fulfillment problem has been extensively studied, and Acimovic and Farias (2019) provide a tutorial on related algorithms developed for addressing the fulfillment optimization problem. In terms of problem context, the most relevant works are those related to omnichannel and multi-item order fulfillment (Xu et al. 2009, Acimovic and Graves 2015, Jasin and Sinha 2015, Wei et al. 2021, Ma 2023, Chen et al. 2024, Zhao et al. 2025). However, these works do not address fulfillment involving a variety of carrier shipping options.

Among the research on order fulfillment, demand uncertainty is the most commonly studied source of randomness (Jasin and Sinha 2015, Das et al. 2023, DeValve et al. 2023, Ma 2023, Zhao et al. 2025). In this paper, demand uncertainty is not considered because the focus is on a single-period order fulfillment problem rather than a forward-looking approach that spans multiple periods.

A few studies do address delivery time uncertainty at various stages of the order fulfillment process or throughout the entire process. For instance, Raj et al. (2024) propose an integrated queuing model that accounts for delivery time uncertainty in different parts of the order life cycle. They incorporate these uncertainties into an optimization model that minimizes costs while adhering to a delivery probability constraint to ensure on-time delivery.

In contrast to using queuing models, this study is particularly relevant to data-driven machine learning-based methods that address delivery time uncertainty in order fulfillment. Specifically, Liu et al. (2021) integrate point predictions of travel time with a last-mile delivery order assignment optimization problem. They propose both a sample average approximation (SAA) model and a distributionally robust optimization model, with the latter constructing a moment-based ambiguity set. Similarly, Kandula et al. (2021) explore a last-mile delivery problem, focusing on delivery success as the uncertainty. They map success probabilities to delivery times, which are then integrated into a vehicle routing problem with time windows. Bayram et al. (2022) study a robust order batching optimization problem in warehouse picking and packing. They use random forest (RF) to predict order processing time to form an uncertainty set within a robust optimization model. This paper builds on similar concepts, developing tractable approximations and reformulations of integer programming-based optimization models. More precisely, the uncertainty in the order fulfillment problem considered in this paper is the potential order delivery deviations, which can occur at various stages of the fulfillment process. A similar uncertainty addressed using a data-driven approach can be found in the work of Salari et al. (2022), who develop a quantile regression forest-based method to generate distributional forecasts of order delivery times. Their approach then applies a decision rule, which is designed to balance the asymmetric costs of early and late deliveries, to produce expected arrival times for customers. This paper differs in two key aspects. First, the distributional forecasts are directly incorporated into contextual optimization models, eliminating the need for a separate decision rule to generate point predictions. Second, the CSO framework proposed in this paper is more comprehensive, accommodating multiple carrier options, whereas Salari et al. (2022) focus on a single-carrier scenario.

3. The Omnichannel Multicourier Order Fulfillment Problem

This section introduces a real-world order fulfillment problem encountered by the industrial partner.

3.1. High-Level Problem Description

This section provides a high-level overview of the online order fulfillment challenges faced by the industrial partner. Similar to other major e-commerce companies, tens of thousands of orders are placed continuously throughout the day on the website and await processing. Each incoming order is organized by stock-keeping units (SKUs), resulting in one or more order lines. For each order line, there are hundreds of fulfillment centers across the company network eligible to source the SKUs. These centers include primarily physical stores as well as regional distribution centers and metro e-commerce centers. Each location is equipped with multiple eligible carriers capable of providing different levels of service, such as ground,

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same-day, and overnight shipping. Throughout this paper, "carrier" refers to one of these specific carrier-service pairs. The set of available carriers spans from major shipping companies to crowdsourced gig carriers. Figure 1 illustrates the order fulfillment life cycle within the operations of the industrial partner.

Before making an order fulfillment decision, the company has access to both the transactional details of the order and the real-time status of the fulfillment network. Specifically, the order information includes the date and time that the order was placed; the type, dimensions, weight, and quantities of the SKUs; the shipping destination; and the desired delivery date. The network status data provide details on the on-hand inventory levels of the requested SKUs and the remaining open capacities at various fulfillment centers. In addition, each carrier provides an estimated table for shipping costs and planned time in transit.

Given this information, the problem is to decide which fulfillment center to source each order line from and which carrier to use for shipping from that location. The goal is to identify the most cost-effective fulfillment option (a location-carrier pair) while ensuring that the customer's desired delivery date is met. The uncertainty of this problem lies in the delivery timeliness performance of each fulfillment option.

3.1.1. Overview of the Current Practice. Currently, the industrial partner uses a fulfillment optimization engine that employs a straightforward greedy algorithm. For each unit in an order, the fulfillment optimization engine sorts the eligible location-carrier pairs by the estimated shipping costs and transit times.

There are several limitations with this approach. First, because the engine makes decisions for each unit independently, it misses the opportunity to consolidate multiple units of the same or different products within an order. This can lead to higher shipping costs as the potential to reduce the number of packages—and thereby, lower shipping expenses—is not fully realized.

Second, the method relies solely on static transit time estimates provided by the carriers based on their service levels to select the best carrier for meeting the promised delivery date. This approach assumes that these estimates are always accurate. However, static transit times may not accurately reflect the actual delivery performance of carriers. Historical data from the industrial partner reveal that over 10% of orders experience varying degrees of early or late deliveries, which are called *deviations* in this paper. Given the complexity of the entire fulfillment process, these deviations can be attributed to numerous factors, such as warehouse pick-and-pack performance, late carrier arrivals, no-shows, and external factors like extreme weather or other unexpected circumstances. Such deviations from customer expectations can negatively impact their experience, potentially harming the company in the long run.

3.2. The Nominal Optimization Problem

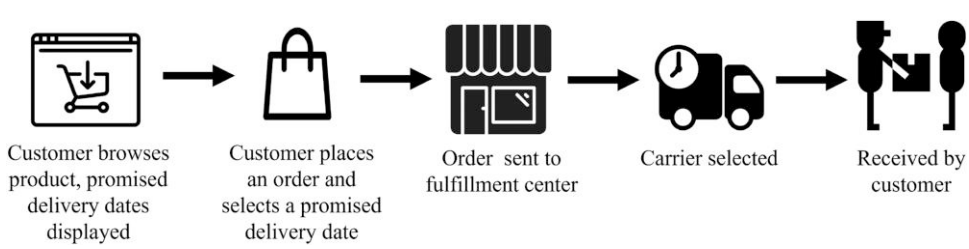
This section introduces the nominal MILP model developed for optimizing the omnichannel multicourier order fulfillment problem. A comprehensive list of the notations presented in this section is provided in Online Appendix A for reference.

3.2.1. Main Decision Variables. Let $\mathcal{I}$ be the set of SKUs and $\mathcal{V}$ be the set of fulfillment options. Here, $\mathcal{V}$ is defined as the Cartesian product $\mathcal{K} \times \mathcal{L}$, where $\mathcal{K}$ denotes the set of carriers and $\mathcal{L}$ denotes the set of locations.

Each order needs to solve an independent optimization problem to decide the selection of carrier-location pairs for each SKU. The primary decision variables are denoted by the vector $\mathbf{z} = (z_{ik\ell})_{i\in\mathcal{I}, k\in\mathcal{K}, \ell\in\mathcal{L}}$, where each $z_{ik\ell} \in \mathbb{Z}$ indicates the quantity of SKU i sourced from location ℓ and shipped by carrier k.

3.2.2. Objective Function. The objective of the optimization problem for an order is to minimize the overall fulfillment costs while ensuring timely delivery. The fulfillment costs for an order include the fixed fulfillment (nonparcel) costs and the shipping (parcel) costs, with consolidation discounts applied when multiple units are assigned to the same carrier-location pair. Delivery timeliness is measured by deviations from the desired delivery date, whether early or late. To encourage timeliness, the objective imposes penalties for both early and late deliveries.

Figure 1. The Life Cycle of the Order Fulfillment Process



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The objective function is formally defined as follows. Let $c^{ship}_{ik\ell} > 0$ be the per-unit shipping cost when using location ℓ and carrier k to fulfill SKU i. For each location ℓ, let $c^{fixed}_{\ell} > 0$ be the per-unit fixed fulfillment cost. For each carrier k, let $\beta_k \in (0,1)$ be the discount percentage applied to shipping costs when multiple units are shipped using carrier k from the same fulfillment center. Let $\gamma^+ \geq 0$ and $\gamma^- \geq 0$ be constants that convert late and early delivery penalties into per-unit costs, respectively.

In an ideal setting where all carrier-location deviations are known at the time of fulfillment, let the vector $\mathbf{d} = (d_{k\ell})_{k\in\mathcal{K},\ell\in\mathcal{L}}$ represent the realized uncertain delivery deviations, where each $d_{k\ell}$ denotes the number of days that the carrier-location pair (k,ℓ) deviates from desired delivery date. A positive deviation indicates late delivery, a zero deviation indicates on-time delivery, and a negative deviation indicates early delivery.

The weighted objective function for an order can be constructed as follows:

$$g(\mathbf{z},\mathbf{d}) = \begin{cases} \sum_{i\in\mathcal{I}}\sum_{k\in\mathcal{K}}\sum_{\ell\in\mathcal{L}}[c^{fixed}_{\ell} + (1-\beta_k)c^{ship}_{ik\ell} + \gamma^+ d^+_{k\ell} + \gamma^- d^-_{k\ell}]z_{ik\ell} & \text{if } \sum_{i\in\mathcal{I}} z_{ik\ell} \geq 2, \\ \sum_{i\in\mathcal{I}}\sum_{k\in\mathcal{K}}\sum_{\ell\in\mathcal{L}}(c^{fixed}_{\ell} + c^{ship}_{ik\ell} + \gamma^+ d^+_{k\ell} + \gamma^- d^-_{k\ell})z_{ik\ell} & \text{otherwise,} \end{cases} \tag{1}$$

where $d^+_{k\ell} = \max\{0, d_{k\ell}\}$ and $d^-_{k\ell} = \max\{0, -d_{k\ell}\}$. This objective function encourages item consolidation within an order by applying a carrier-specific discount if at least two units are sourced from the same location and shipped by the same carrier. In addition, it leverages an asymmetric penalty cost function for delivery deviations, where $\gamma^+ \gg \gamma^-$. This asymmetry strongly penalizes late deliveries to discourage delays while also applying a smaller penalty to disincentivize early deliveries. Specifically, the penalties can be interpreted as the opportunity cost associated with early deliveries and the lost-sale cost associated with late deliveries. The objective function is piecewise linear and can be linearized using standard rewritings.

3.2.3. Operational Constraints. Several operational constraints ensure that the selection of fulfillment center locations and carriers is feasible in practice. For location $\ell \in \mathcal{L}$, let $\text{inv}_{i\ell} \geq 0$ be the inventory level of SKU i at location ℓ and $\text{cap}_{\ell} \geq 0$ be the available capacity at location ℓ when the order is placed. Let $q_i \geq 0$ be the quantity of SKU i in the order, and let $e_{ik\ell} \in \{0,1\}$ be a binary indicator of whether the location-carrier pair (ℓ,k) is an eligible fulfillment option for SKU i. An assignment $\mathbf{z}$ in the feasible set $\mathcal{Z}$ must obey the following constraints.

- Each unit of an SKU must be sourced from exactly one location and shipped by exactly one carrier:

$$\sum_{k\in\mathcal{K}}\sum_{\ell\in\mathcal{L}} z_{ik\ell} = q_i, \quad \forall i \in \mathcal{I}. \tag{2}$$

- Each SKU can only be fulfilled by eligible candidate location-carrier pairs, which may vary depending on the carrier's coverage and package requirements:

$$z_{ik\ell} \leq q_i e_{ik\ell}, \quad \forall i \in \mathcal{I}, k \in \mathcal{K}, \ell \in \mathcal{L}. \tag{3}$$

- Only locations with available inventory of the requested SKUs can be chosen:

$$\sum_{k\in\mathcal{K}} z_{ik\ell} \leq \text{inv}_{i\ell}, \quad \forall i \in \mathcal{I}, \ell \in \mathcal{L}. \tag{4}$$

- Each location has a daily capacity limit on the total number of units that it can process:

$$\sum_{i\in\mathcal{I}}\sum_{k\in\mathcal{K}} z_{ik\ell} \leq \text{cap}_{\ell}, \quad \forall \ell \in \mathcal{L}. \tag{5}$$

3.3. The Contextual Stochastic Order Fulfillment Problem

In practice, the delivery deviations of each carrier-location pair cannot be perfectly estimated in advance. To account for this uncertainty for each order, this section models the vector of uncertain delivery deviations $\tilde{\mathbf{d}} = (\tilde{d}_{k\ell})_{k\in\mathcal{K},\ell\in\mathcal{L}} \sim \mathbb{P}$ as a $(|\mathcal{K}||\mathcal{L}|)$-dimensional random vector governed by a joint probability distribution $\mathbb{P}$. Because deviations measured in days only take on discrete values in the data set, each $\tilde{d}_{k\ell}$ inherently follows a discrete probability distribution. It is further assumed that every $\tilde{d}_{k\ell}$ shares the same finite, ordered support. To formalize this, let C be the total number of distinct deviation values and denote them by $\xi_1 < \xi_2 < \cdots < \xi_C$. By definition, any realization $d_{k\ell}$ of $\tilde{d}_{k\ell}$ must lie in this set: $d_{k\ell} \in \{\xi_1, \xi_2, \ldots, \xi_C\}$.

In addition, let $\mathbf{s} = (\mathbf{s}_{k\ell})_{k\in\mathcal{K},\ell\in\mathcal{L}}$ denote the state vector observable to the order before the fulfillment decision is made, where each $\mathbf{s}_{k\ell}$ is a vector of covariates encompassing information about order details and fulfillment network status related to carrier-location pair (k,ℓ). The contextual distribution of the deviation vector, upon observing $\mathbf{s}$, is denoted by $\mathbb{P}(\mathbf{d}|\mathbf{s})$, and the individual deviation contextual distribution is $\mathbb{P}_{k\ell}(d_{k\ell}|\mathbf{s}_{k\ell})$ for each $k \in \mathcal{K}$ and $\ell \in \mathcal{L}$.

This leads to the following contextual stochastic order fulfillment problem (CSOFP):

$$\text{(CSOFP)} \quad \min_{\mathbf{z}\in\mathcal{Z}} \mathbb{E}_{\tilde{\mathbf{d}}\sim\mathbb{P}(\mathbf{d}|\mathbf{s})}[g(\mathbf{z},\tilde{\mathbf{d}})]. \tag{6}$$

Solving Problem (6) in practice requires leveraging data from past orders. Unlike classical CSO formulations where the entire random cost vector is observed, CSOFP is based on an observational data set of past decisions and their realized costs. Let $\mathcal{O}$ be the set of historical orders. For each $o \in \mathcal{O}$, define the decision

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vector $\mathbf{z}_o = (z_{ik\ell,o})_{i\in\mathcal{I},k\in\mathcal{K},\ell\in\mathcal{L}}$. Because deviations were observed only for chosen (i,k,ℓ) triples, the full deviation vector $\mathbf{d}_o = (d_{k\ell,o})_{k\in\mathcal{K},\ell\in\mathcal{L}}$ contains many unobserved entries. The subset of realized deviations for order o is

$$\mathbf{D}_o := \left\{ d_{k\ell,o} \,\middle|\, \sum_{i\in\mathcal{I}} z_{ik\ell,o} > 0, k\in\mathcal{K}, \ell\in\mathcal{L} \right\}.$$

The resulting data set available for estimating the contextual distribution $\mathbb{P}(\mathbf{d}|\mathbf{s})$ is $\mathbb{D} = \{(\mathbf{s}_o, \mathbf{D}_o)\}_{o=1}^{|\mathcal{O}|}$.

4. Methodology: A CSO Framework for Observational Data

Solving CSOFP under observational data requires (1) learning a contextual distributional oracle from partial realizations and (2) optimizing with that oracle in a tractable way.

This section lays out a contextual stochastic optimization framework that does exactly that. In Section 4.1, a generic CSO problem (denoted by $\mathcal{P}$) is defined. Section 4.2 shows how to construct a contextual distribution oracle $\mathcal{M}(\mathbf{s})$ from partial observations. Sections 4.3 and 4.4 introduce two solution paradigms—contextual sample average approximation and contextual robust optimization—which use $\mathcal{M}(\mathbf{s})$ as input. Finally, Section 4.6 maps all of the above back to the omnichannel multicourier CSOFP introduced in Section 3. Figure 2 provides a schematic overview of the proposed framework, illustrating the flow from historical observational data through the learning and optimization stages to produce a final decision.

4.1. Generic Contextual Stochastic Optimization Problem

Problem $\mathcal{P}$ consists of fulfilling a set of incoming requests sequentially, each of which specifies demands for a subset of products. Let $\mathcal{I}$ be the set of products, $\mathcal{V}$ be the set of selection options (e.g., fulfillment or transport modes), and $\mathcal{Z} \subseteq \mathbb{Z}_+^{|\mathcal{I}|\times|\mathcal{V}|}$ be the feasible assignment set. Each request arrives with known demand quantities $\{q_i : i\in\mathcal{I}\}$ at decision time. The decision variable $\mathbf{z} = (z_{iv})_{i\in\mathcal{I},v\in\mathcal{V}} \in \mathcal{Z}$ assigns quantity z_{iv} of product i to option v.

Let $\tilde{\mathbf{c}} = (\tilde{c}_{iv})_{i\in\mathcal{I},v\in\mathcal{V}} \sim \mathbb{P}$ denote the random cost vector. Assume that $\tilde{\mathbf{c}}$ is element-wise positive and that the cost function $g(\mathbf{z},\tilde{\mathbf{c}})$ is piecewise linear in $\mathbf{z}$. Introduce for $j = 0,1,\ldots,n$ the break-point vectors

$$\mathbf{b}^j = (b_{iv}^j)_{i\in\mathcal{I},v\in\mathcal{V}},\ \mathbf{b}^0 < \mathbf{b}^1 < \cdots < \mathbf{b}^n \quad \text{(component wise)}$$

so that each segment is defined as

$$[\mathbf{b}^{j-1}, \mathbf{b}^j] = \{\mathbf{z} : b_{iv}^{j-1} \le z_{iv} \le b_{iv}^j \,\forall i\in\mathcal{I}, v\in\mathcal{V}\}.$$

Let $\mathbf{a}^j = (a_{iv}^j)_{i\in\mathcal{I},v\in\mathcal{V}}$ be the marginal-cost adjustment on segment j. Then,

$$g(\mathbf{z},\tilde{\mathbf{c}}) = (\tilde{\mathbf{c}} + \mathbf{a}^j)^\top \mathbf{z} \quad \text{if} \quad \mathbf{z}\in[\mathbf{b}^{j-1},\mathbf{b}^j],\ j = 1,\ldots,n, \tag{7}$$

thereby capturing volume-dependent pricing behaviors.

For each request, the fulfillment decision solves the risk-neutral stochastic optimization problem

$$\min_{\mathbf{z}\in\mathcal{Z}} \mathbb{E}_{\tilde{\mathbf{c}}\sim\mathbb{P}}\,[g(\mathbf{z},\tilde{\mathbf{c}})]. \tag{8}$$

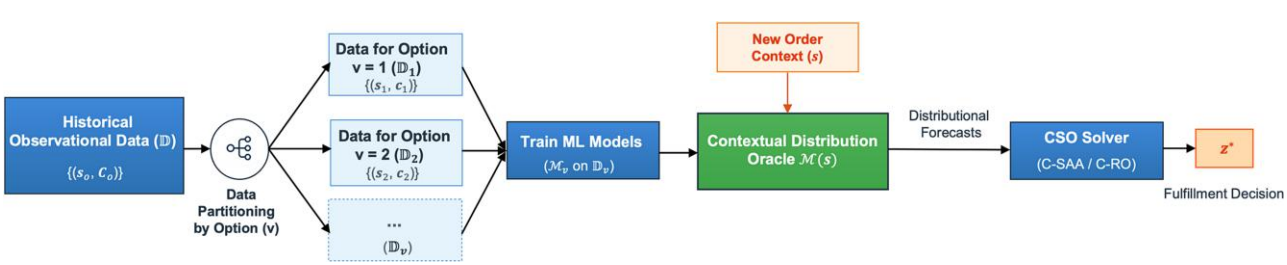
The piecewise-linear cost in (7) partitions the decision space into shipment-size tiers determined by the break points $\{\mathbf{b}^j\}$. Within tier j, the marginal cost is $\tilde{\mathbf{c}} + \mathbf{a}^j$, which shifts to $\tilde{\mathbf{c}} + \mathbf{a}^{j+1}$ upon entering tier $j+1$. This formulation accommodates economies of scale (declining unit costs at higher volumes), quantity discounts (step-down pricing beyond thresholds), capacity surcharges (unit-cost premiums when exceeding lower-cost bands), and other tiered-pricing schemes.

In Problem (8), the true distribution $\mathbb{P}$ is not readily available. However, side information often exists for each product-option pair. Let $\mathbf{s}_{iv}$ denote the covariate vector for product i under selection option v (e.g., network state, weather, or current demand). All of these covariate vectors are aggregated into the state vector $\mathbf{s} = (\mathbf{s}_{iv})_{i\in\mathcal{I},v\in\mathcal{V}}$, which encompasses all contextual information available at the time of decision. The distribution $\mathbb{P}$ is then approximated by the conditional distribution $\mathbb{P}(\mathbf{c}|\mathbf{s})$, yielding the contextual stochastic optimization problem

$$\min_{\mathbf{z}\in\mathcal{Z}} \mathbb{E}_{\tilde{\mathbf{c}}\sim\mathbb{P}(\mathbf{c}|\mathbf{s})}[g(\mathbf{z},\tilde{\mathbf{c}})]. \tag{9}$$

Let $\tilde{\mathbf{c}}_v = (\tilde{c}_{iv})_{i\in\mathcal{I}}$ and $\mathbf{s}_v = (\mathbf{s}_{iv})_{i\in\mathcal{I}}$ for each $v\in\mathcal{V}$. Under the assumption that conditional on its own

Figure 2. (Color online) A Schematic Overview of the CSO Framework for Observational Data



Note. ML stands for machine learning.

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covariates, each component c_v is independent across options $v \in \mathcal{V}$, one has

$$\mathbb{P}(\mathbf{c}|\mathbf{s}) = \Pi_{v\in\mathcal{V}}\mathbb{P}_v(\mathbf{c}_v|\mathbf{s}_v),$$

where $\mathbb{P}_v(\cdot|\mathbf{s}_v)$ is the contextual distribution for option v. This assumption is reasonable when each option's cost depends primarily on its own covariates. Shared features, such as weather, create similarity but do not induce direct dependence. Consequently, Problem (9) can be written as

$$\min_{\mathbf{z}\in\mathcal{Z}} \mathbb{E}_{\tilde{\mathbf{c}}\sim\Pi_{v\in\mathcal{V}}\mathbb{P}_v(\mathbf{c}_v|\mathbf{s}_v)}[g(\mathbf{z},\tilde{\mathbf{c}})]. \tag{10}$$

4.2. Constructing a Contextual Distribution Oracle from Observational Data

The contextual distribution $\mathbb{P}(\mathbf{c}|\mathbf{s})$ is also generally unknown but can often be inferred from historical observations. In many real-world CSO applications (including CSOFP), however, data are observational (i.e., only the costs corresponding to decisions actually taken were recorded). As a result, the data set available to estimate $\mathbb{P}(\mathbf{c}|\mathbf{s})$ takes the form $\mathbb{D} = \{(\mathbf{s}_o, \mathbf{C}_o)\}_{o=1}^{|\mathbb{D}|}$, where $\mathbf{s}_o = (s_{iv,o})_{i\in\mathcal{I},\, v\in\mathcal{V}}$ and $\mathbf{C}_o = (\mathbf{c}_{v,o})_{v\in\mathcal{V}}$ collect the context and realized costs, with $\mathbf{c}_{v,o}$ observed only if option v was selected in request o.

The learning task consists of finding a full vector forecasting oracle $\mathcal{M}$ that approximates $\mathbb{P}(\mathbf{c}|\mathbf{s})$ in a supervised manner using the data set $\mathbb{D}$; $\mathcal{M}$ receives a context $\mathbf{s}$ as input and returns a probability distribution $\mathcal{M}(\mathbf{s})$ on the cost vector $\tilde{\mathbf{c}}$. Under the conditional independence assumption from the previous section, learning $\mathbb{P}(\mathbf{c}|\mathbf{s})$ amounts to learning $\mathbb{P}_v(\mathbf{c}_v|\mathbf{s}_v)$ for each selection option v separately. Partition the full data set $\mathbb{D}$ into subsets $\mathbb{D}_v = \{(\mathbf{s}_{v,1}, \mathbf{c}_{v,1}), (\mathbf{s}_{v,2}, \mathbf{c}_{v,2}), \ldots\}$, representing the data set for option v. A machine learning model $\mathcal{M}_v$ is then trained on $\mathbb{D}_v$ to approximate each $\mathbb{P}_v(\mathbf{c}_v|\mathbf{s}_v)$. Assemble the oracle as the product $\mathcal{M}(\mathbf{s}) = (\mathcal{M}_v(\mathbf{s}_v))_{v\in\mathcal{V}}$, which generates joint forecasts by treating each component independently. Note that if every $\mathbf{s}_v$ shares the same structure and dimension, a single learning architecture can serve as $\mathcal{M}_v$ for all $v \in \mathcal{V}$, trained on the union $\cup_{v\in\mathcal{V}}\mathbb{D}_v$.

By reducing the high-dimensional joint learning task to $|\mathcal{V}|$ manageable subproblems, this decomposition turns observational data into a black-box contextual distribution oracle $\mathcal{M}(\mathbf{s})$, which will serve as the input for the solution methods developed in Section 4.3 (C-SAA) and Section 4.4 (C-RO).

4.3. Contextual Sample Average Approximation

Problem (10) is approximated by generalizing the SAA method (Kleywegt et al. 2002) to contextual stochastic optimization. Given a request with covariates $\mathbf{s}$, the contextual sample average approximation proceeds in two stages.

1. Candidate generation.
   a. Draw N_1 independent cost vector samples $\{\boldsymbol{\xi}^n\}_{n=1}^{N_1} \sim \mathcal{M}(\mathbf{s})$, where $\boldsymbol{\xi}^n = (\xi_{iv}^n)_{i\in\mathcal{I},\, v\in\mathcal{V}}$.
   b. Solve

$$\mathbf{z}_i = \arg\min_{\mathbf{z}\in\mathcal{Z}} \frac{1}{N_1}\sum_{n=1}^{N_1} g(\mathbf{z}, \boldsymbol{\xi}^n), \quad i = 1, \ldots, Q,$$

   independently Q times to obtain candidate solutions $\{\mathbf{z}_1, \ldots, \mathbf{z}_Q\}$ with sample average objectives $\{f_1, \ldots, f_Q\}$.
   c. Compute the average objective $\underline{f} = \frac{1}{Q}\sum_{i=1}^{Q} f_i$, which is a statistical lower bound on the true optimal value f^* because $\underline{f}$ is an unbiased estimator of $\mathbb{E}_{\tilde{\mathbf{c}}\sim\mathbb{P}(\mathbf{c}|\mathbf{s})}[\underline{f}]$ and $\mathbb{E}_{\tilde{\mathbf{c}}\sim\mathbb{P}(\mathbf{c}|\mathbf{s})}[\underline{f}] \le f^*$.
2. Candidate evaluation.
   a. For each $\mathbf{z}_i$, draw a larger set of evaluation samples $\{\boldsymbol{\xi}^n\}_{n=1}^{N_2} \sim \mathcal{M}(\mathbf{s})$, with $N_2 \gg N_1$.
   b. Compute the re-estimated objectives $\overline{f}(\mathbf{z}_i) = \frac{1}{N_2}\sum_{n=1}^{N_2} g(\mathbf{z}_i, \boldsymbol{\xi}^n)$.
   c. Each $\overline{f}(\mathbf{z}_i)$ is an upper bound on f^* because each candidate solution is feasible; the difference $\overline{f}(\mathbf{z}_i) - \underline{f}$ approximates the optimality gap.
   d. Select the candidate $\mathbf{z}^*$ with the smallest estimated gap.

4.3.1. Discussion. Unlike the reweighting approach of Bertsimas and Kallus (2020), which leverages complete historical samples for each fully observed cost vector, C-SAA relies solely on observational data (i.e., only the costs of the selected options are available). Attempting to form all possible cost vector combinations from partial observations would lead to a combinatorial explosion as $|\mathcal{V}|$ and $|\mathbb{D}|$ grow, making such a method intractable in large, high-dimensional selection spaces.

4.4. Contextual Robust Optimization

The performance of C-SAA largely hinges on the accuracy of the estimated contextual distribution. Robust optimization provides a hedge against misspecification of the contextual distribution by guarding against worst-case scenarios, making it a viable approach for contextual stochastic optimization. Let $\mathcal{U}(\mathbf{s}) \subset \mathbb{R}^{|\mathcal{I}|\times|\mathcal{V}|}$ represent a contextual uncertainty set for the cost vector $\tilde{\mathbf{c}}$. The contextual robust optimization problem is

$$\min_{\mathbf{z}\in\mathcal{Z}} \max_{\boldsymbol{\xi}\in\mathcal{U}(\mathbf{s})} g(\mathbf{z}, \boldsymbol{\xi}). \tag{11}$$

Two constructions of $\mathcal{U}(\mathbf{s})$ are considered: a discrete sampling-based set $\mathcal{U}_d(\mathbf{s})$ and a budgeted interval set $\mathcal{U}_b(\mathbf{s}, B)$.

4.4.1. Contextual Discrete Uncertainty Set. Problem (11) can be reformulated into a single-level problem by enumerating every possible realization of the random

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vector $\tilde{\mathbf{c}}$, a standard technique in robust optimization. In principle, a discrete distribution can be generated as mentioned before by combining all combinations of the observed data points. However, this will make the problem intractable as discussed.

To retain tractability, a sampling-based discrete uncertainty set in the same spirit of C-SAA is introduced. Let $\mathcal{U}_d(\mathbf{s}) = \{\boldsymbol{\xi}^1, \ldots, \boldsymbol{\xi}^N\}$ denote a collection of N deviation scenarios drawn from $\mathcal{M}(\mathbf{s})$. The robust counterpart of (11) becomes

$$\text{(C-RO-D)} \quad \min_{\mathbf{z}\in\mathcal{Z},\, t\in\mathbb{R}} \quad t \qquad \text{s.t.} \quad t \geq g(\mathbf{z}, \boldsymbol{\xi}^n), \quad n = 1, \ldots, N. \tag{12}$$

4.4.2. Contextual Budgeted Interval Uncertainty Set.

An alternative uncertainty set is the budgeted interval set $\mathcal{U}_b(\mathbf{s}, B)$ (Bertsimas and Sim 2004), which is defined by a robust budget B. This budget limits how many components of the uncertain parameter $\tilde{\mathbf{c}}$ can simultaneously deviate to their upper bounds.

Rather than estimating the full joint distribution, only the conditional lower and upper quantiles at levels q_{low} and q_{up} are needed to define a $(q_{\text{low}}, q_{\text{up}})$ prediction interval for each option.

For each option $v \in \mathcal{V}$, let $\underline{\boldsymbol{\xi}}_v = (\underline{\xi}_{iv})_{i\in\mathcal{I}}$ and $\overline{\boldsymbol{\xi}}_v = (\overline{\xi}_{iv})_{i\in\mathcal{I}}$ denote the estimated conditional q_{low} and q_{up} quantiles extracted from $\mathcal{M}_v(\mathbf{s}_v)$:

$$\underline{\boldsymbol{\xi}}_v = \inf\{\mathbf{u} : \mathbb{P}_v(\mathbf{c}_v \leq \mathbf{u} | \mathbf{s}_v) \geq q_{\text{low}}\},$$
$$\overline{\boldsymbol{\xi}}_v = \inf\{\mathbf{u} : \mathbb{P}_v(\mathbf{c}_v \leq \mathbf{u} | \mathbf{s}_v) \geq q_{\text{up}}\}.$$

Given a budget B, the budgeted interval uncertainty set is

$$\mathcal{U}_b(\mathbf{s}, B) = \left\{ \boldsymbol{\xi} : \xi_{iv} = \underline{\xi}_{iv} + \delta_{iv}(\overline{\xi}_{iv} - \underline{\xi}_{iv}), \sum_{i,v} \delta_{iv} \leq B,\ \delta_{iv} \in [0,1] \right\}.$$

With a fixed $\mathbf{z}$, the inner maximization problem in (11) is

$$\max_{\delta_{iv}} \quad \sum_{i,v} (\overline{\xi}_{iv} - \underline{\xi}_{iv}) z_{iv} \delta_{iv} \tag{13}$$

$$\text{s.t.} \quad \sum_{i,v} \delta_{iv} \leq B \tag{14}$$

$$\delta_{iv} \in [0,1], \quad \forall i, v. \tag{15}$$

Let $y_j \in \{0,1\}$ and $w^j_{iv} \geq 0$ be auxiliary variables. Denote by $\pi \geq 0$ and $\lambda_{iv} \geq 0$ the dual variables corresponding to Constraints (14) and (15), respectively. Finally, let M be a sufficiently large constant. A valid choice for the "big-M" constant is $M = \max_{i,v,j}\{b^n_{iv} - b^1_{iv}, b^{n-1}_{iv}, (\underline{\xi}_{iv} + a^j_{iv}) b^n_{iv}\}$. By taking the dual of the inner maximization problem and introducing y_j and w^j_{iv}, a robust counterpart of (11) is obtained as

$$\begin{aligned}
\text{(C-RO-B)} \quad \min_{\mathbf{z}\in\mathcal{Z}, y, w, \pi, \lambda} \quad & \left(\sum_{i,v,j} w^j_{iv} + \pi B + \sum_{i,v} \lambda_{iv} \right) \\
\text{s.t.} \quad & \pi + \lambda_{iv} \geq (\overline{\xi}_{iv} - \underline{\xi}_{iv}) z_{iv}, \quad \forall i, v \\
& z_{iv} \leq b^j_{iv} + M(1 - y_j), \quad \forall i, v, j \\
& z_{iv} \geq b^{j-1}_{iv} - M(1 - y_j), \quad \forall i, v, j \\
& w^j_{iv} \geq (\underline{\xi}_{iv} + a^j_{iv}) z_{iv} - M(1 - y_j), \quad \forall i, v, j \\
& y_j \in \{0,1\}, w^j_{iv} \geq 0, \pi \geq 0, \lambda_{iv} \geq 0.
\end{aligned} \tag{16}$$

4.5. Performance Guarantees on C-SAA and C-RO

C-SAA replaces the unknown cost distribution with Monte Carlo samples generated by a learned contextual oracle and minimizes the resulting empirical objective. Under standard regularity conditions, the SAA objective is statistically consistent; its optimal value converges exponentially fast to the true optimum as N_1 grows (Kleywegt et al. 2002, Shapiro et al. 2021). When the oracle is well calibrated and N_1 is large, C-SAA thus attains the smallest *expected* cost. However, it remains sensitive to distributional misspecification and finite-sample error, which can expose the solution to rare but severe tail events.

C-RO mitigates this risk by optimizing the worst-case cost over a prescribed uncertainty set. Let a feasible solution $\mathbf{z}$ satisfy $\max_{\boldsymbol{\xi}\in\mathcal{U}(\mathbf{s})} g(\mathbf{z}, \boldsymbol{\xi}) \leq t$. For a discrete uncertainty set built from N samples (C-RO-D), one obtains with probability $1 - \delta$ that

$$\Pr(g(\mathbf{z}, \tilde{\mathbf{c}}) > t) \leq \epsilon$$

provided

$$N \geq 2|\mathcal{V}||\mathcal{I}| + \frac{2|\mathcal{V}||\mathcal{I}|}{\epsilon} \ln\left(\frac{2}{\epsilon}\right) + \frac{2}{\epsilon} \ln\left(\frac{1}{\delta}\right)$$

samples are drawn (Calafiore and Campi 2006, Campi and Garatti 2008, Bertsimas et al. 2021). Although this bound grows linearly in the dimension of the uncertainty $|\mathcal{V}||\mathcal{I}|$, it too can degrade under distributional misspecification. Under a budgeted interval uncertainty set (C-RO-B), the violation probability instead satisfies

$$\Pr(g(\mathbf{z}, \tilde{\mathbf{c}}) > t) \leq \exp\left(-\frac{B^2}{2|\mathcal{V}||\mathcal{I}|}\right)$$

so that the budget parameter B directly controls the trade-off between average-case performance and robustness (Bertsimas and Sim 2004). This tunable flexibility makes C-RO-B an attractive middle ground between C-SAA and C-RO-D.

Extending these guarantees to bound worst-case regret or postdecision surprise under partial-feedback settings—where counterfactual costs are unobserved—remains an important direction for future work.

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4.6. Application to the Omnichannel Multicourier CSOFP

The CSO framework applies directly to CSOFP via the following identifications:

$$\begin{aligned} &\mathcal{I} \longmapsto \{\text{SKUs}\}, && \mathcal{V} \longmapsto \mathcal{K} \times \mathcal{L}, \\ &z_{iv} \longmapsto z_{ik\ell}, && \mathbf{s}_v \longmapsto \mathbf{s}_{k\ell}, \\ &\tilde{\mathbf{c}} = (\tilde{c}_v)_{v\in\mathcal{V}} \longmapsto \tilde{\mathbf{d}} = (\tilde{d}_{k\ell})_{k\in\mathcal{K},\,\ell\in\mathcal{L}}. && \mathbb{P}_v \longmapsto \mathbb{P}_{k\ell}. \end{aligned}$$

Under the conditional independence assumption,

$$\mathbb{P}(\mathbf{d}\,|\,\mathbf{s}) = \prod_{k\in\mathcal{K}} \prod_{\ell\in\mathcal{L}} \mathbb{P}_{k\ell}(d_{k\ell}\,|\,\mathbf{s}_{k\ell}).$$

Hence, CSOFP (Problem (6)) can be rewritten as

$$\min_{\mathbf{z}\in\mathcal{Z}} \mathbb{E}_{\tilde{\mathbf{d}}\sim\Pi_{k,\ell}\mathbb{P}_{k\ell}(d_{k\ell}\,|\,\mathbf{s}_{k\ell})}[g(\mathbf{z},\tilde{\mathbf{d}})]. \tag{17}$$

To learn each marginal distribution $\mathbb{P}_{k\ell}(d_{k\ell}|\mathbf{s}_{k\ell})$, the historical data set is first partitioned into $\mathbb{D}_{k\ell} = \{(\mathbf{s}_{k\ell,m}, d_{k\ell,m})\}_{m=1}^{M_{k\ell}}$. This is followed by training a per-carrier model $\mathcal{M}_k$ on the data set $\mathbb{D}_k = \cup_{\ell\in\mathcal{L}}\mathbb{D}_{k\ell}$. This per-carrier approach is a practical adaptation of the generic per-option framework from Section 4.2. Rather than training a distinct model for each individual location-carrier pair (which can be inefficient and suffer from data sparsity), the data are pooled to train a single more robust model for each carrier. This decomposition simplifies learning, allows the model to generalize to new locations, and was found to outperform a single global model in preliminary experiments. The full contextual distribution oracle is then assembled as $\mathcal{M}(\mathbf{s}) = (\mathcal{M}_k(\mathbf{s}_{k\ell}))_{k\in\mathcal{K},\,\ell\in\mathcal{L}}$.

4.6.1. Applying C-SAA and C-RO. Computing the expectation in $\mathbb{E}_{\tilde{\mathbf{d}}\sim\mathcal{M}(\mathbf{s})}[g(\mathbf{z},\tilde{\mathbf{d}})]$ directly would require either enumerating all $C^{|\mathcal{K}|\times|\mathcal{L}|}$ deviation vectors or numerically integrating a continuous cumulative distribution function (CDF)—both computationally intractable. Instead, the two solution paradigms use the contextual distribution oracle $\mathcal{M}(\mathbf{s})$ as follows.

- C-SAA replaces the expectation by a sample average; draw N_1 full vector scenarios $\{\tilde{\mathbf{d}}^n\} \sim \mathcal{M}(\mathbf{s})$, and solve

$$\min_{\mathbf{z}\in\mathcal{Z}} \frac{1}{N_1}\sum_{n=1}^{N_1} g(\mathbf{z},\tilde{\mathbf{d}}^n).$$

- C-RO builds an uncertainty set from oracle outputs—either the discrete scenario set $\{\tilde{\mathbf{d}}^n\}_{n\in[N]}$ or a budgeted interval set from marginal quantiles—and then, solves

$$\min_{\mathbf{z}\in\mathcal{Z}} \max_{\mathbf{d}\in\mathcal{U}(\mathbf{s})} g(\mathbf{z},\mathbf{d}).$$

Both approaches avoid the exponential support growth while remaining agnostic to the internal structure of $\mathcal{M}$. A full C-RO-B reformulation is provided in Online Appendix F.

5. Learning the Contextual Distribution of Delivery Deviations for CSOFP

Having defined in Section 4 a generic contextual distribution oracle $\mathcal{M}(\mathbf{s})$, this section describes its concrete implementation for CSOFP. It begins by selecting features to construct the covariate vector. Two classes of machine learning methods are then proposed to estimate the contextual distribution of deviations: (1) probabilistic multiclass classification, which leverages the discrete support of deviations, and (2) tree-based quantile regression, which directly estimates the conditional CDF and avoids quantile crossing (Salari et al. 2022).

For simplicity, the carrier index k is omitted in the notation, and the data set for training the per-carrier model is written as $\mathbb{D} = \{(\mathbf{s}_m, d_m)\}_{m\in[M]}$.

5.1. Contextual Feature Selection

Carefully selecting contextual features is crucial for constructing informed covariates for delivery deviation predictions as various stages in the order life cycle can contribute to deviations.

Candidate features fall into three categories.

1. *Order level*. This includes customer location, SKU weight and dimensions, SKU type, quantities, release hour, release day, and planned lead time (i.e., the difference between the promised delivery date and the order release time). These factors can influence deviations through logistical complexity, handling requirements, and time-sensitive elements affecting the order's processing and delivery schedule.

2. *Fulfillment center*. This includes coordinates, location type, on-hand capacity, and on-hand inventory. The geographical location and type of fulfillment center can affect the efficiency of order processing, whereas capacity and inventory levels determine the speed and reliability of fulfilling orders.

3. *Carrier related*. This includes service level, promised transit time, shipping charges, and carrier zone. These factors impact the carrier ability to meet delivery deadlines, with service level and transit times directly affecting deviations, whereas shipping charges and zones reflect logistical challenges and potential bottlenecks.

5.2. Probabilistic Multiclass Classification

This discrete nature allows the contextual distribution learning problem to be framed as a multiclass classification (MC-CLF) problem. Under MC-CLF, each class represents a possible delivery deviation value in number of days, and the output class probabilities correspond directly to the probability mass function of the discrete distribution. For a new sample with covariates vector $\mathbf{s}$, let the conditional probability of class c be $p_c(\mathbf{s}) = \mathbb{P}(d = \xi_c|\mathbf{s})$, with the corresponding predicted probability from MC-CLF denoted as $\hat{p}_c(\mathbf{s})$.

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Multinomial logistic regression (MLR) and classification tree-based models are considered to solve MC-CLF. Specifically, MLR is a classic classifier that aims to estimate the class probabilities by maximizing the log-likelihood function, the negative of the log loss function. The log loss is defined as $-\sum_{m=1}^{M}\sum_{c=1}^{C}\mathbf{1}\{d_m = \xi_c\}\log(\hat{p}_c(\mathbf{s}_m))$. On the other hand, classification trees use a nonparametric approach that recursively partitions the data into subsets based on feature values in a top-down manner, resulting in a tree-like model of decisions.

Let $\mathcal{R}_1, \ldots, \mathcal{R}_R$ be the regions partitioned by a classification tree. The classification tree estimates

$$\hat{p}_c(\mathbf{s}) = \frac{\sum_{r=1}^{R}\mathbf{1}\{\mathbf{s}\in\mathcal{R}_r\}\cdot\mathbf{1}\{d=\xi_c\}}{\sum_{r=1}^{R}\mathbf{1}\{\mathbf{s}\in\mathcal{R}_r\}}, \tag{18}$$

To enhance predictive power, ensemble learning methods, such as bagging and gradient boosting, are employed through the use of RFs and gradient boosted trees (GBTs). The details for their calculation of the class probabilities are available in Online Appendix G.

The performances of the MC-CLF models are evaluated by log loss and Brier score. The latter is defined as the average squared error between predicted probabilities and actual outcomes $\sum_{m=1}^{M}\sum_{c=1}^{C}(\mathbf{1}\{d_m = \xi_c\}\hat{p}_c(\mathbf{s}_m))^2$. For both metrics, lower values indicate better probabilistic prediction performance.

5.2.1. Probability Calibration. Raw class probability estimates $\hat{p}_c(\mathbf{s})$ from classifiers often exhibit miscalibration; a predicted probability α does not correspond to an empirical frequency α (Niculescu-Mizil and Caruana 2005). *Good calibration* means that over all samples with $\hat{p}_c(\mathbf{s}) = \alpha$, the true fraction belonging to class c is approximately α. Formally, let $f(\alpha) = \mathbb{P}(d = \xi_c \mid \hat{p}_c(\mathbf{s}) = \alpha)$ be the calibration function. Perfect calibration means $f(\alpha) = \alpha$ for all $\alpha \in [0,1]$.

Isotonic regression (Zadrozny and Elkan 2002) learns a nondecreasing mapping $g : [0,1] \to [0,1]$ that best aligns raw scores to observed frequencies, preserving the rank order of $\hat{p}_c$. Given a calibration set $\mathbb{D}_{cal} = \{(\hat{p}_c(\mathbf{s}_m), y_{c,m})\}_{m=1}^{M_{cal}}$ with $y_{c,m} = \mathbf{1}\{d_m = \xi_c\}$, isotonic regression solves

$$\min_{g} \sum_{m=1}^{M_{cal}} (y_{c,m} - g(\hat{p}_c(\mathbf{s}_m)))^2$$
$$\text{s.t. } g(\alpha) \le g(\alpha') \text{ whenever } \alpha \le \alpha'.$$

The calibrated probabilities are $\tilde{p}_c(\mathbf{s}) = g(\hat{p}_c(\mathbf{s}))$. By enforcing monotonicity, isotonic calibration ensures that higher raw scores remain higher after mapping, and empirical frequencies within each score bin match the calibrated values.

In this paper, a k-fold crossvalidation procedure is employed; for each fold i, the classifier is trained on its training split, raw scores are obtained on the held-out calibration split, and an isotonic regression model $g^{(i)}$ is fit to those scores. On unseen test instances with covariates $\mathbf{s}$, each fold's classifier and corresponding $g^{(i)}$ yield calibrated probabilities $\tilde{p}_c^{(i)}(\mathbf{s})$, which are then averaged across all k folds to produce the final estimate $\tilde{p}_c(\mathbf{s}) = \frac{1}{k}\sum_{i=1}^{k}\tilde{p}_c^{(i)}(\mathbf{s})$. Because per-class calibration does not guarantee that the final probabilities will sum to one, a normalization step is applied. The vector of calibrated class probabilities $(\tilde{p}_c(\mathbf{s}))_{c=1}^{C}$ is divided by its sum to yield a valid multiclass probability distribution.

5.3. Tree-Based Quantile Regression

Unlike multiclass classification methods that treat each deviation value as an unordered category, quantile regression exploits the inherent ordering of deviations by framing the task as ordinal regression and directly estimating the conditional cumulative distribution function $F(u \mid \mathbf{s}) = \mathbb{P}(d \le u \mid \mathbf{s})$.

Standard QR models minimize, for each quantile level q, the pinball loss

$$\rho_q(d, \hat{d}) = q\max(d - \hat{d}, 0) + (1-q)\max(\hat{d} - d, 0), \tag{19}$$

where $\hat{d}$ denotes the model's predicted qth quantile. Collecting these quantile estimates across q yields a stepwise approximation of the CDF. However, estimating each quantile independently can violate the ordering $\hat{d}_{(q_1)} \le \hat{d}_{(q_2)}$ for $q_1 < q_2$, resulting in a nonmonotonic—and thus, invalid—CDF.

Tree-based QR methods—namely regression trees extended to CDF estimation and their ensemble version quantile random forests (QRFs) (Meinshausen and Ridgeway 2006)—avoid this "quantile crossing" problem by extracting quantiles from the empirical distribution of training samples in each leaf region.

A regression tree partitions the covariate space into disjoint regions to minimize mean squared error. Although regression trees are often used for point forecasts (e.g., predicting the mean response in each leaf), they can be adapted for distributional forecasting by preserving all observations within each partitioned region. For a new sample with covariates $\mathbf{s}$, let $\mathcal{R}(\mathbf{s})$ be the leaf region into which $\mathbf{s}$ falls. The conditional CDF of the deviation can then be approximated by the empirical distribution of the training samples in that leaf:

$$\hat{F}(u \mid \mathbf{s}) = \sum_{m=1}^{M} \frac{\mathbf{1}\{\mathbf{s}_m \in \mathcal{R}(\mathbf{s})\}}{|\{j : \mathbf{s}_j \in \mathcal{R}(\mathbf{s})\}|}\mathbf{1}\{d_m \le u\}. \tag{20}$$

A subsequent kernel density estimation applied to $\{d_m : \mathbf{s}_m \in \mathcal{R}(\mathbf{s})\}$ can further smooth this step-function CDF (Salari et al. 2022).

QRF extends this approach by pooling the leaf-level samples across an ensemble of T trees. For each tree t,

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let $\mathcal{R}^{(t)}(\mathbf{s})$ be leaf region containing $\mathbf{s}$. The QRF estimate of the conditional CDF is

$$\hat{F}(u|\mathbf{s}) = \frac{1}{T}\sum_{t=1}^{T}\sum_{m=1}^{M}\frac{\mathbf{1}\{\mathbf{s}_m \in \mathcal{R}^{(t)}(\mathbf{s})\}}{|\{j : \mathbf{s}_j \in \mathcal{R}^{(t)}(\mathbf{s})\}|}\mathbf{1}\{d_m \le u\}. \tag{21}$$

Because this procedure pools ordered observations directly, it naturally preserves the ordinal support and never produces crossing quantiles.

To measure the distributional forecasting performance, the paper uses two metrics—the continuous ranked probability score (CRPS) and pinball loss. CRPS evaluates the closeness of the estimated CDF $\hat{F}$ given covariates $\mathbf{s}$ to the observed ground-truth value d:

$$\mathrm{CRPS}(\hat{F}, d, \mathbf{s}) = \int_{-\infty}^{\infty}(\hat{F}(u|\mathbf{s}) - \mathbf{1}\{u \ge d\})^2 du. \tag{22}$$

Let ξ and ξ' be two independent random variables with distribution $\hat{F}(u|\mathbf{s})$. The CRPS can alternatively be expressed as

$$\mathrm{CRPS}(\hat{F}, d, \mathbf{s}) = \mathbb{E}(|\xi - d|) - \frac{1}{2}\mathbb{E}(|\xi - \xi'|). \tag{23}$$

This equation allows for an approximation of CRPS by sampling from $\hat{F}(u|\mathbf{s})$. Pinball loss measures the accuracy of quantile predictions by penalizing errors asymmetrically depending on whether the predictions are overestimates or underestimates.

6. Case Study

This section presents computational results that demonstrate the potential of the proposed CSO framework when evaluated on real-world industrial instances.

6.1. Data Overview

The industrial partner provides a month-long data set in 2023, covering more than 1 million online orders for home delivery and over 20,000 unique SKUs. Each order can be sourced from roughly 250–1,000 possible fulfillment centers across the company's network and shipped via 1 of 10–30 distinct carriers.

The target variable in the machine learning problem, delivery deviation, is defined as the difference between the actual delivered date and the promised delivery date. To enhance the quality of the analysis, order lines with extreme deviations were excluded—specifically, those with absolute values greater than 10 days. Such outliers often reflect exceptional circumstances, including returns to sender, lost shipments, or atypical business orders, which can introduce noise and bias into the model.

6.2. Evaluation of Contextual Distributional Learning of Delivery Deviations

This section evaluates the contextual distributional learning models presented in Section 5 for the case study.

6.2.1. Machine Learning Model Settings. For the evaluation of the learning models, the data set was partitioned into a training set comprising orders from the first three weeks (approximately 1.3 million order lines) and a test set consisting of orders from the final week (approximately 0.4 million order lines). For multi-SKU orders, features were aggregated by retaining only those of the SKU with the highest shipping charge and quantity. The MC-CLF models were implemented using scikit-learn (Pedregosa et al. 2011) and CatBoost (Prokhorenkova et al. 2018), with probability calibration applied to the best-performing uncalibrated models based on crossvalidated metrics. Quantile regression models, including regression tree and QRF, were implemented via the quantile-forest package (Johnson 2024). All hyperparameters were tuned using threefold crossvalidation in Optuna (Akiba et al. 2019); MC-CLF models were tuned to minimize log loss, whereas quantile regression models were tuned to minimize CRPS.

6.2.2. Learning Performance Analysis. In Table 1, two standard metrics—log loss and Brier score—are reported to measure the accuracy of probabilistic predictions for each tuned MC-CLF model on the test set. Calibration yields a clear improvement; the calibrated classifier achieves the lowest log loss (0.965) and Brier score (0.031), down from 1.050 and 0.034, respectively, for the uncalibrated GBT (GBT-CLF), substantially outperforming both the RF classifier (RF-CLF; 1.593 and 0.054, respectively) and the simpler baselines (MLR and classification tree).

Figure 3 presents the joint predicted distributions of the Calibrated-classification (CLF) (Figure 3(a)), the QRF (Figure 3(b)), and the discretized regression tree (Figure 3(c)), each plotted against the true empirical distribution for orders fulfilled by a representative carrier. The Calibrated-CLF closely reproduces the empirical mass at zero deviation (on-time deliveries) and tightly follows the tails corresponding to early and late deliveries. The QRF exhibits slightly greater discrepancies, particularly in the majority classes (−1 to 1), whereas the discretized regression tree shows the largest misalignment from the empirical distribution.

Table 2 reports pinball losses at selected quantiles and CRPS values for the tuned quantile regression and Calibrated-CLF models. CRPS was approximated by drawing 1,000 samples to compute Equation (23).

Table 1. Out-of-Sample Average Log Loss and Brier Score Across All Carriers

Model	MLR	Classification tree	RF-CLF	GBT-CLF	Calibrated-CLF
Log loss	1.950	2.242	1.593	1.050	**0.965**
Brier score	0.073	0.072	0.054	0.034	**0.031**

Note. The best model for each metric is highlighted in bold.

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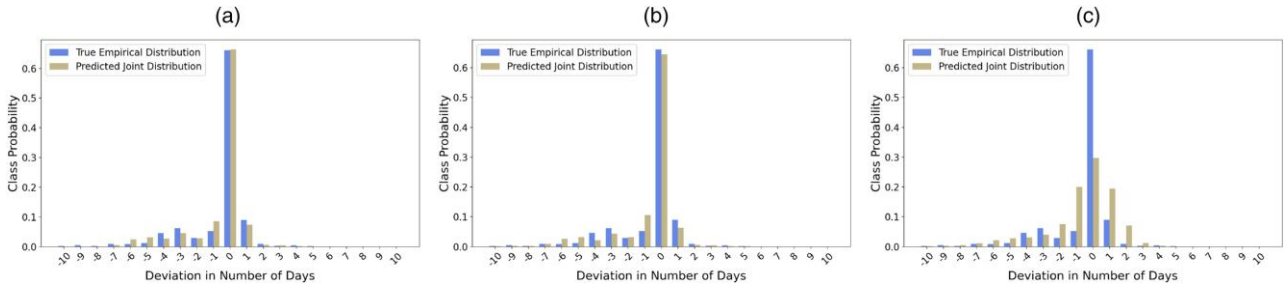
Figure 3. (Color online) Out-of-Sample Comparison of the Joint Predicted Distribution and True Empirical Distribution for an Example Carrier



Notes. (a) Calibrated-CLF. (b) QRF. (c) Regression tree (discretized).

The Calibrated-CLF model exhibit higher pinball losses than the two quantile regression models, reflecting its inherently discrete output, but nonetheless, it achieves the lowest CRPS overall, even though it was not directly tuned for this metric. This superior CRPS performance likely results from calibration producing sharper probability estimates, especially for the majority deviation classes.

Additionally, Figure 4 displays the 95% prediction intervals created by the three contextual distribution learning methods for an example carrier. The visualization aligns with the pinball loss results, showing that QRF generally provides the narrowest interval widths. Although the intervals generated by the Calibrated-CLF cover most of the observed values, their coverage is less effective at the two extremes compared with the two QR methods. The inferior performance of the Calibrated-CLF at extreme quantiles is likely because of the discretization of the output space, which limits precision, and the cumulative probability method, which can cause jumps when selecting quantiles. In contrast, regression tree and QRF provide finer-grained predictions, leading to more accurate and narrower intervals at the extremes.

In summary, MC-CLF models excel at providing distributional forecasts, whereas QR models are better suited to generating prediction intervals on this data set. Because QRF benefits from ensembling multiple regression trees to reduce variance and improve accuracy and consistently outperforms a single regression tree, only QRF is kept for QR in the subsequent CSO evaluations.

Table 2. Out-of-Sample Average Pinball Loss and CRPS Across All Carriers

	Pinball loss					
Model	$q_{0.025}$	$q_{0.05}$	$q_{0.5}$	$q_{0.95}$	$q_{0.975}$	CRPS
Calibrated-CLF	0.199	0.198	0.676	0.526	0.304	**0.386**
QRF	**0.033**	**0.057**	**0.263**	**0.143**	0.096	0.402
Regression tree	0.064	0.109	0.302	0.147	**0.093**	0.515

Note. The best model for each metric is highlighted in bold.

6.3. Baseline Models

To compare with the proposed contextual optimization models, four baseline order fulfillment algorithms are introduced below.

1. *Pure cost-driven greedy heuristic (Greedy)*. Mimics the industrial partner's fulfillment optimization engine by selecting, for each unit of an SKU, the carrier-location pair that minimizes the fulfillment cost. Formally, for each order, it solves $\min_{\mathbf{z}\in\mathcal{Z}}\{\sum_{i,k,\ell}(c_\ell^{fixed}+c_{ik\ell}^{ship})z_{ik\ell}\}$.

2. *Empirical-SAA*. Follows the SAA framework of Section 4.3 except that each deviation sample is drawn from the overall empirical distribution P^E of the training set without using any contextual information. For each $n\in[N_1]$ and every $(k,\ell)\in\mathcal{K}\times\mathcal{L}$, $\xi_{k\ell}^n\sim P^E$, $\boldsymbol{\xi}^n=(\xi_{k\ell}^n)_{k\in\mathcal{K},\,\ell\in\mathcal{L}}$. The empirical SAA problem is $\min_{z\in\mathcal{Z}}\frac{1}{N_1}\sum_{n=1}^{N_1}g(\boldsymbol{z},\boldsymbol{\xi}^n)$.

3. *C-Empirical-SAA*. A slightly smarter SAA baseline that maintains separate empirical distributions P_k^E for each carrier. For each n and (k,ℓ), $\xi_{k\ell}^n\sim P_k^E$, $\boldsymbol{\xi}^n=(\xi_{k\ell}^n)_{k\in\mathcal{K},\,\ell\in\mathcal{L}}$, and it solves the same SAA problem as above.

4. *Point predict-then-optimize (PTO)*. Trains per-carrier regression models $h_k(\cdot)$ to estimate the conditional mean deviations for each $(k,\ell)\in\mathcal{K}\times\mathcal{L}$, producing the point prediction vector $\hat{\mathbf{d}}=\{h_k(\mathbf{s}_{k\ell})\}_{k\in\mathcal{K},\ell\in\mathcal{L}}$. It then solves the deterministic problem $\min_{z\in\mathcal{Z}}g(z,\hat{\mathbf{d}})$. This method is effective only if the objective function $g(\mathbf{z},\tilde{\mathbf{d}})$ is linear in $\tilde{\mathbf{d}}$, in which case estimating the conditional distribution simplifies to estimating the conditional mean. This is because by linearity of expectation, $\mathbb{E}_{\tilde{\mathbf{d}}\sim\mathbb{P}(\mathbf{d}\,|\,\mathbf{s})}[g(\mathbf{z}_o,\tilde{\mathbf{d}})]=g(\mathbf{z},\mathbb{E}_{\tilde{\mathbf{d}}\sim\mathbb{P}(\mathbf{d}\,|\,\mathbf{s})}[\tilde{\mathbf{d}}])$. However, $g(z,\tilde{\mathbf{d}})$ is nonlinear because of asymmetric penalty costs of the deviations; hence, replacing the distribution by its mean may bias the solution. For each carrier, the regression model with the lowest crossvalidated mean squared error is selected (see Online Appendix H for details).

6.4. Experimental Setting

This section outlines the computational environment, simulation framework, and parameter configurations used in the experiments.

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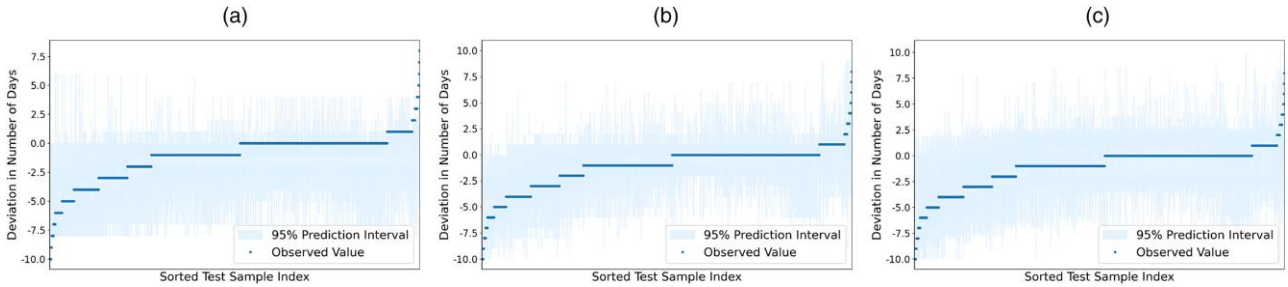
Figure 4. (Color online) Sorted Observed Deviation Values vs. 95% Prediction Intervals for an Example Carrier on Out-of-Sample Data



Notes. (a) Calibrated-CLF prediction intervals. (b) QRF prediction intervals. (c) Regression tree prediction intervals.

6.4.1. Implementation Environment. The optimization models in this study were implemented in Python 3.9 using Gurobi 11.0.0, with eight threads allocated to each order instance. A termination criterion of a 1% mixed-integer programming (MIP) gap or a five-minute time limit was applied. All experiments were conducted on a server equipped with dual-socket Intel Xeon Gold 6226 CPUs, each featuring 24 cores running at 2.7 GHz.

6.4.2. Simulation Framework. One of the difficulties in this application is that the ground truth is not available for options that were not selected. To overcome this limitation, a simulation environment was built to generate synthetic "ground truth" when an algorithm selects an option. Specifically, delivery deviations were drawn from a multinomial distribution, with probabilities estimated by fitting a fine-tuned, calibrated multi-class classification model based on orders from the last week of the data set (testing period). Obviously, these data were not accessible to the algorithm and only used by the simulator to replicate real-world conditions. The training and calibration procedures of the simulator closely followed the methodology outlined in Section 5.2. An evaluation of the simulator is provided in Online Appendix J.

For the following experiments, as mentioned in Section 6.2, all machine learning models were trained on order data from the first three weeks. Evaluation was conducted on 5,000 multi-item orders randomly sampled from the final week. To account for variability, solution metrics were computed across 50 random realizations. *All reported metrics were normalized by setting the Greedy baseline to one for each random seed; true performance values were omitted because of data confidentiality.*

6.4.3. Model Names and Parameters. In the following, the C-SAA problems using samples from the contextual distribution estimates provided by the MC-CLF and QRF models are referred to as C-SAA-CLF and C-SAA-QRF, respectively. Similarly, the C-RO problems are referred to as C-RO-{B,D}-CLF and C-RO-{B,D}-QRF, respectively.

By default, a constant $\beta_k = 0.5$ consolidation discount factor for all $k \in \mathcal{K}$ was considered. The default penalty for early delivery γ^- was 0.1, and that for late delivery γ^+ was 10. For SAA-based methods, the default was to have $Q = 20$ independent replications, each with a sample size of $N_1 = 30$. The obtained solutions were then evaluated with a default sample size of $N_2 = 300$. The default sample size used for C-RO-D was $N = 200$. As for C-RO-B, a default of a 95% interval was considered. Moreover, a scaling factor $\eta \in [0,1]$ (default of 0.5) determined the uncertainty budget in C-RO-B defined as $B = \eta|\mathcal{K}||\mathcal{L}|$.

6.5. Overall Results

Table 3 reports the normalized average and worst-case objective values across 50 random seeds and all order instances. Based on these results, the following observations are drawn.

6.5.1. Average Performance. The C-SAA variants and C-RO-B-CLF method lead to more reduction in the expected objective than other methods. C-SAA-CLF achieves the greatest reduction, with a normalized mean of 0.820 ± 0.003 (–18.0% versus Greedy) followed by C-SAA-QRF at 0.844 ± 0.003 (–15.6%) and C-RO-B-CLF at 0.842 ± 0.003 (–15.8%). By comparison, Empirical-SAA (0.962 ± 0.002) and PTO (0.878 ± 0.003) offer smaller improvements, and C-Empirical-SAA (0.870 ± 0.003) performs only marginally better than PTO.

6.5.2. Worst-Case Performance. The C-RO-D methods deliver the strongest guarantees under the worst simulated outcome; C-RO-D-CLF attains 0.779 (–22.1%), C-RO-D-QRF attains 0.787 (–21.3%), and C-RO-B-CLF attains 0.811 (–18.9%). These three methods uniformly dominate the baseline and other approaches in the worst-case metric.

Overall, the C-SAA approaches excel in minimizing the expected objective, whereas the C-RO-D variants excel in worst-case robustness. Notably, C-RO-B-CLF strikes a strong balance between average efficiency and tail-risk protection.

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Table 3. Average and Worst-Case Realized Objective Values Across Different Order Fulfillment Methods

Method	Average objective (95% CI)	Worst-case objective
Greedy	1.000 ± 0.000	1.000
PTO	0.878 ± 0.003	0.919
Empirical-SAA	0.962 ± 0.002	1.049
C-Empirical-SAA	0.870 ± 0.003	0.896
C-SAA-CLF	**0.820 ± 0.003**	0.823
C-SAA-QRF	**0.844 ± 0.003**	0.842
C-RO-B-CLF	**0.842 ± 0.003**	**0.811**
C-RO-B-QRF	0.861 ± 0.003	0.824
C-RO-D-CLF	0.928 ± 0.003	**0.779**
C-RO-D-QRF	0.914 ± 0.003	**0.787**

Notes. The top three methods for each metric are highlighted in bold. CI stands for confidence interval.

6.6. Effectiveness of C-SAA

This section further compares the performance of the C-SAA methods with the two Empirical-SAA baselines and the PTO.

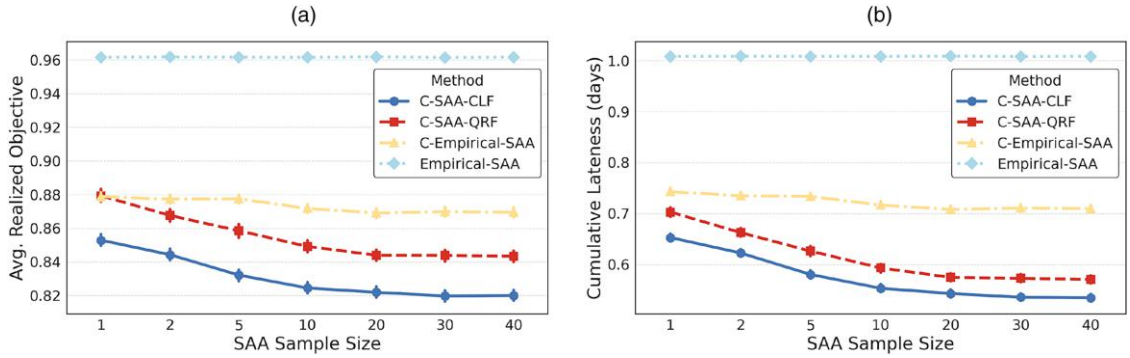
6.6.1. Value of Contextual Information. To facilitate a closer examination of the performance of the SAA-based models, Figure 5 shows the impact of varying sample sizes (N_1). On the one hand, larger sample sizes should provide a more accurate approximation of the expected value in Problem (6), assuming that the delivery deviation distributions were well estimated. On the other hand, it is important to maintain a reasonably small sample size to ensure computational efficiency. For both C-SAA methods, the effect of increasing sample sizes is obvious. The average objectives are reduced from 0.85 to 0.82 for C-SAA-CLF and from 0.88 to 0.84 for C-SAA-QRF, indicating their increasing alignment with the true risk-neutral objective function. This improvement is particularly evident in their enhanced effectiveness at avoiding cumulative lateness, defined as the total number of late delivery days across all units in the orders. The cumulative lateness decreases by 14% for C-SAA-CLF and by 12% for C-SAA-QRF.

In contrast, both Empirical-SAA and C-Empirical-SAA struggle to improve their performances in both the average realized objective value and the cumulative lateness as the sample size grows. This is because relying solely on the empirical distribution fails to capture the variability in delivery deviation distributions, which may depend on contextual information specific to each order. Biased distributional forecasts thus guide the model wrongly to create solutions with inferior quality. Based on these observations, both contextual distributional forecasting methods proposed in the paper are successful in this application, demonstrating a stronger performance compared with using empirical distribution for delivery deviation forecasting. In summary, the value of contextual information is particularly important in order fulfillment decision making, where the uncertainty of delivery deviation can vary drastically across different states of the environment.

6.6.2. Value of Distributional Forecasts. This section examines the advantage of employing distributional forecasting models over the simpler and more common point prediction model within the CSO framework. As supported by theory, simply using the conditional mean estimate of deviations to replace the expected value in CSOFP can introduce bias when approximating the true objective value, thus degrading solution quality. Figure 6 confirms that under changing γ^+, CSO methods with distributional estimator (C-SAA-CLF and C-SAA-QRF) consistently outperform PTO in both average realized objective value (Figure 6(a)) and cumulative lateness (Figure 6(b)). In fact, the performance differences become more pronounced as the late delivery penalty (γ^+) increases.

When γ^+ rises from 2.5 to 40, PTO achieves only a 7% reduction in cumulative lateness, whereas C-SAA-CLF and C-SAA-QRF realize reductions of approximately

Figure 5. (Color online) C-SAA vs. Empirical SAA Across Different Sample Sizes



Notes. (a) Performance in average (Avg.) realized objective value. (b) Performance in cumulative lateness.

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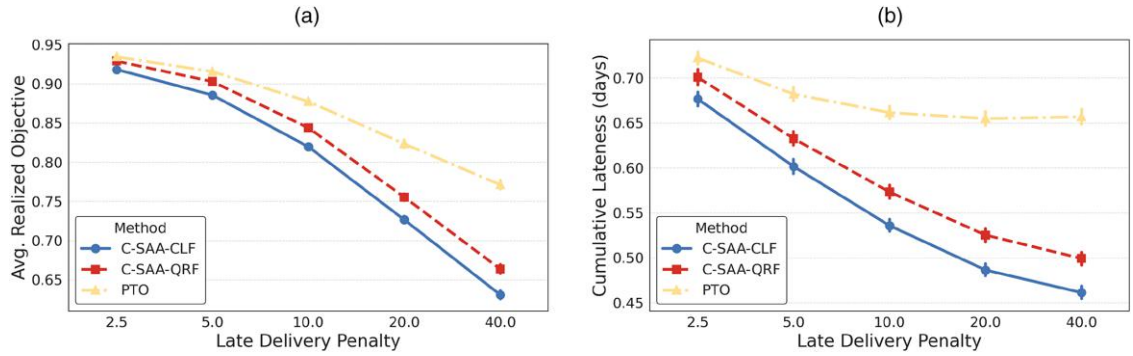
Figure 6. (Color online) C-SAA vs. PTO Across Different Late Delivery Penalties



Notes. (a) Performance in average (Avg.) realized objective value. (b) Performance in cumulative lateness.

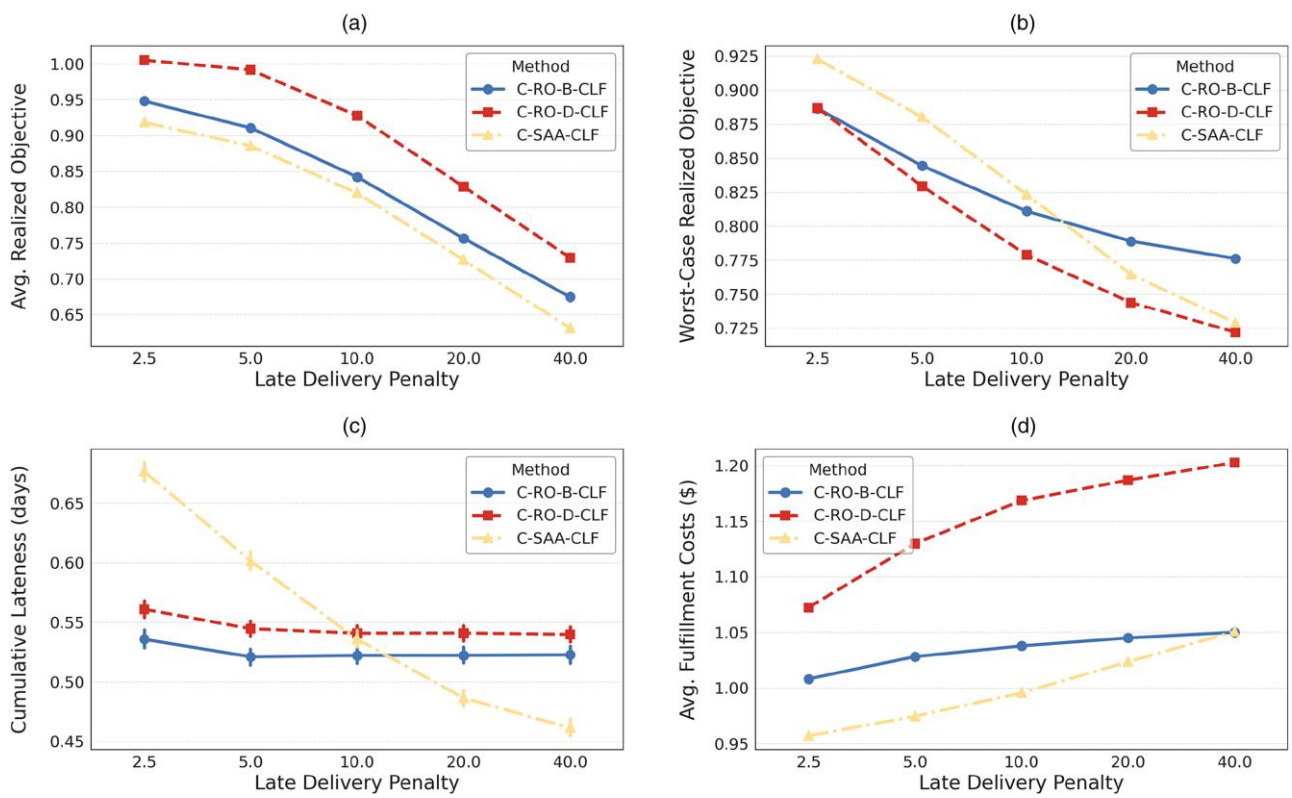
22% and 20%, respectively. These results demonstrate that distributional forecasting not only enables lower overall lateness but also leverages higher penalties more effectively, yielding a sharper decline in average objective and making C-SAA particularly well suited to settings where delivery timeliness is critical.

6.7. Cost of Robustness

To quantify the trade-off between robustness and efficiency, C-RO-B-CLF and C-RO-D-CLF were compared with C-SAA-CLF as the late delivery penalty (γ^+) increases (Figure 7).

As depicted in Figure 7(a), at $\gamma^+ = 2.5$, average objectives are approximately 0.92 for C-SAA-CLF, 0.95 for C-RO-B-CLF, and 1.00 for C-RO-D-CLF. As γ^+ rises to 40, these values decline in parallel—C-SAA-CLF to 0.63, C-RO-B-CLF to 0.67, and C-RO-D-CLF to 0.73. Throughout this range, C-SAA-CLF consistently achieves the lowest average objective followed by C-RO-B-CLF and then C-RO-D-CLF.

Figure 7. (Color online) C-RO vs. C-SAA Across Different Late Delivery Penalties



Notes. (a) Performance in average (Avg.) realized objective value. (b) Performance in worst-case realized objective value. (c) Performance in cumulative lateness. (d) Performance in fulfillment costs.

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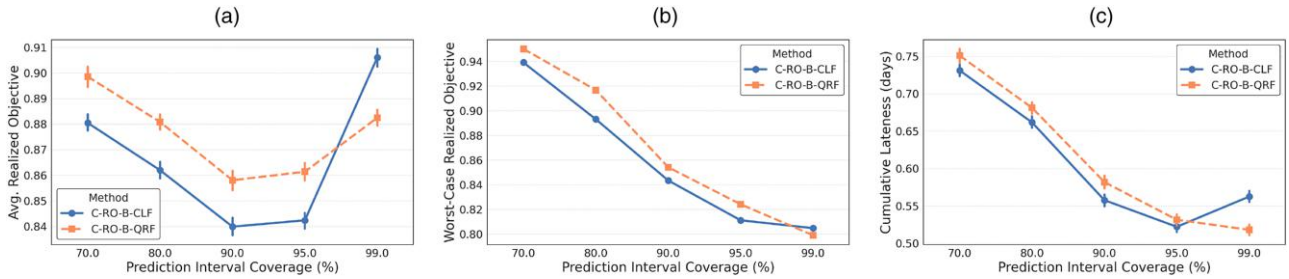
Figure 8. (Color online) Performance of C-RO-B Across Different Prediction Interval Coverages



Notes. (a) Performance in average (Avg.) objective (Obj). (b) Performance in worst-case Obj. (c) Performance in cumulative lateness.

Worst-case objectives (Figure 7(b)) underscore the robustness of the C-RO methods; C-RO-D-CLF consistently yields the strongest tail protection, dropping from 0.88 at $\gamma^+ = 2.5$ to 0.72 at $\gamma^+ = 40$. C-RO-B-CLF falls from 0.885 to 0.775, whereas C-SAA-CLF declines more steeply from 0.925 to 0.73. Notably, at $\gamma^+ = 40$, C-SAA-CLF nearly matches C-RO-D-CLF, demonstrating that C-SAA can approach robust performance under extreme penalty scenarios.

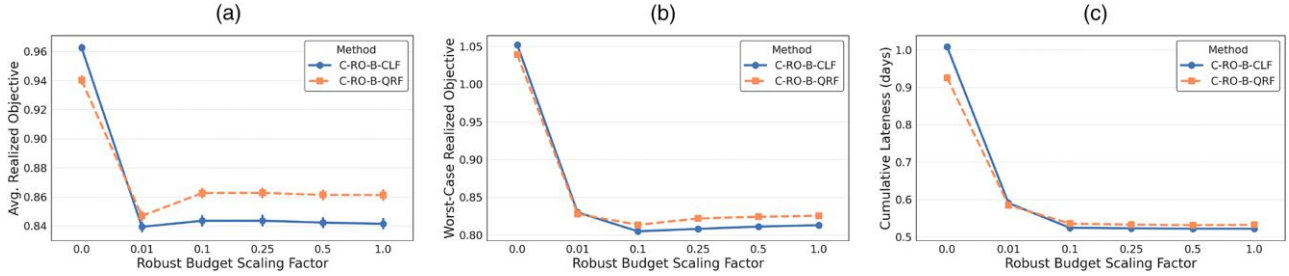
This robustness manifests in cumulative lateness (Figure 7(c)). Both C-RO variants tightly control lateness with minimal sensitivity to γ^+, whereas C-SAA-CLF's lateness drops substantially from 0.67 at $\gamma^+ = 2.5$ to 0.46 at $\gamma^+ = 40$. These robustness gains incur higher expected fulfillment costs (Figure 7(d)). C-RO-D-CLF's average cost increases from 1.07 to 1.20, C-RO-B-CLF's average cost increases from 1.01 to 1.05, and C-SAA-CLF's average cost increases from 0.96 to 1.05. Thus, although all methods incur higher costs under stricter penalties, C-RO-B-CLF exhibits the smallest relative cost increase, and C-SAA-CLF and C-RO-D-CLF have more aggressive increases in costs because of more conservative decisions.

A final strategic consideration lies in selecting γ^+ to reflect firm priorities. Lower γ^+ emphasizes immediate cost savings—maximizing short-term profits—whereas higher γ^+ prioritizes on-time delivery, fostering customer satisfaction, loyalty, and long-term value. The CSOFP framework encodes this trade-off directly via γ^+; robust formulations (C-RO-B-CLF and C-RO-D-CLF) further reduce the need for precise γ^+ tuning by delivering stable, worst-case performance across penalty levels, whereas C-SAA-CLF can leverage high γ^+ most aggressively when strict timeliness is paramount.

6.7.1. Flexibility of C-RO-B. The C-RO models effectively hedge against worst-case delivery deviations and potential misspecifications in the prediction models. However, this robustness incurs more conservative decisions and higher fulfillment costs. The C-RO-B variants retain the distribution-free guarantees while introducing two tuning parameters—prediction interval coverage and robust uncertainty budget—that directly control conservativeness. Lower coverage levels and smaller budgets yield less conservative, more cost-efficient solutions (at the expense of risk protection), whereas higher values emphasize hedging against extreme deviations by sacrificing average performance.

Figures 8 and 9 summarize how these parameters affect solution quality under both C-RO-B-CLF and C-RO-B-QRF as the prediction interval coverage increases from 70% to 90%. In Figure 8(a), increasing coverage from 70% to 90% reduces the average objectives to about 0.84 (C-RO-B-CLF) and 0.86 (C-RO-B-QRF), an approximately 4% improvement relative to 70% coverage. This is consistent with the concurrent decline in cumulative lateness shown in Figure 8(c). However, pushing coverage beyond 95% reverses this trend; at 99% coverage, the average objective rises sharply because of the cost of extreme conservatism

Figure 9. (Color online) Performance of C-RO-B Across Different Robust Uncertainty Budgets



Notes. (a) Performance in average (Avg.) objective (Obj). (b) Performance in worst-case Obj. (c) Performance in cumulative lateness.

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and for C-RO-B-CLF, degraded prediction accuracy (reflected in a spike in lateness). By contrast, C-RO-B-QRF continues to reduce lateness at 99%, illustrating its superior tail modeling. In terms of worst-case performance (Figure 8(b)), both methods improve with coverage, but C-RO-B-CLF plateaus after 95%, whereas C-RO-B-QRF continues to improve at 99%.

The effect of the robust budget, controlled by the scaling factor η, appears in Figure 9. Increasing η from 0 to 0.01 leads to a steep drop in the average objective for both methods (Figure 9(a)). Both methods reach their minimum values at $\eta = 0.01$. At the same time, the worst-case objective follows a similar pattern (Figure 9(b)); it falls sharply up to $\eta = 0.1$ and then, edges back up for both methods as the budget grows. This trend is also seen in cumulative lateness, which drops dramatically until $\eta = 0.1$ and then levels off (Figure 9(c)). Together, these trends show that moderate budgets capture most of the gains in both average- and worst-case performances, whereas further increases deliver diminishing returns.

6.8. Impact of Consolidation Factor

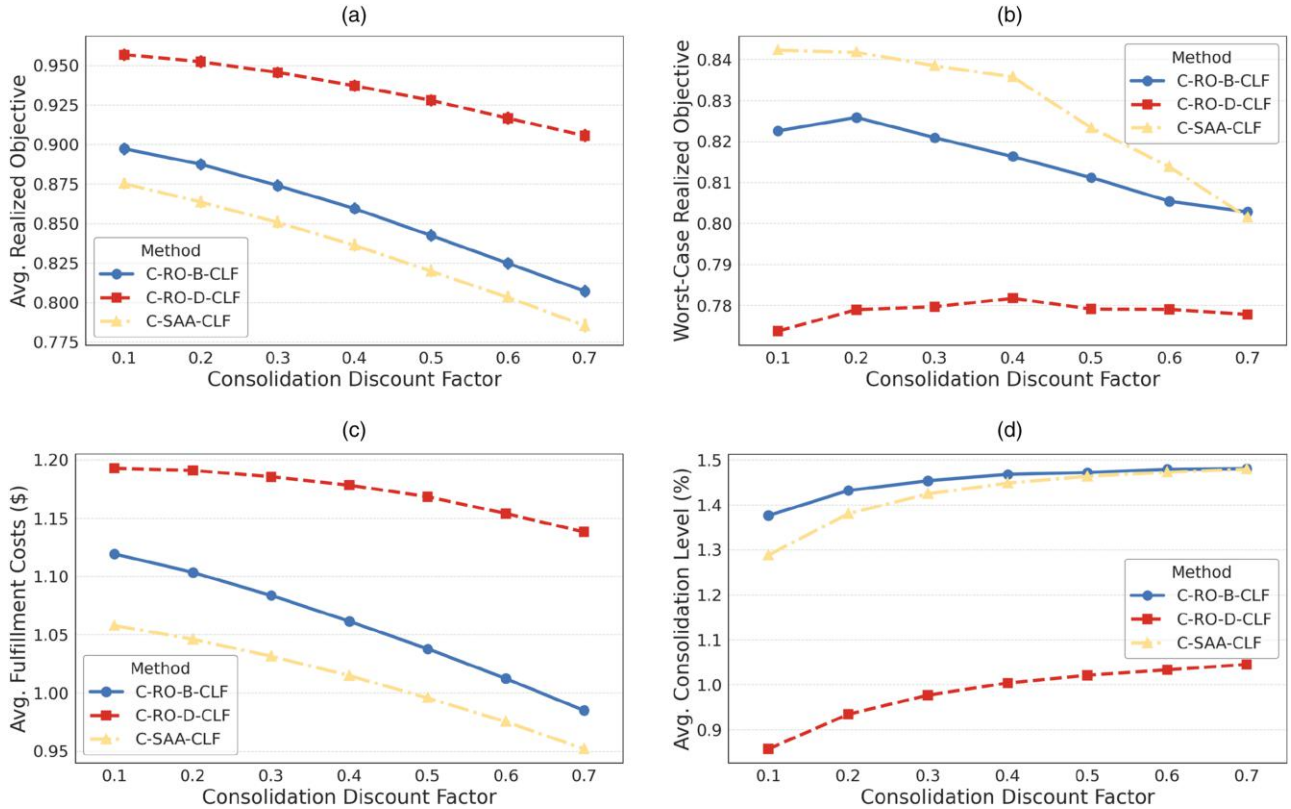
This section provides a discussion on the impact of item consolidation. CSOFP relies on the coefficients β_k to control the incentive to batch items. The industrial partner does not know the exact percentage of discount to be offered for item consolidation, which may depend on a variety of convoluted factors. Therefore, this section performs sensitivity analysis on a range of possible discount factors (from 0.1 to 0.7).

Figure 10 compares the performance of three leading methods—C-SAA-CLF, C-RO-B-CLF, and C-RO-D-CLF—across average and worst-case realized objective, fulfillment cost, and consolidation level. Consolidation level is defined as the percentage of units batched (sourced and shipped by the same location-carrier pair) in each order.

As shown in Figure 10(a), deeper discounts lead each method to exploit batching incentives more aggressively. C-SAA-CLF consistently attains the lowest average objective across all discount factors, declining from approximately 0.875 at $\beta = 0.1$ to about 0.785 at $\beta = 0.7$. C-RO-B-CLF follows closely, from 0.895 down to 0.805, whereas C-RO-D-CLF remains the most conservative, decreasing from 0.955 to 0.905.

The worst-case objective exhibits a more differentiating pattern (Figure 10(a)). Both C-RO-D-CLF and C-RO-B-CLF are less susceptible to change in the discount, whereas C-SAA-CLF starts at 0.842 and falls to 0.802 under high discounts.

Figure 10. (Color online) Performance of CSO Methods Across Different Consolidation Discount Factors



Notes. (a) Performance in average (Avg.) realized objective value. (b) Performance in worst-case realized objective value. (c) Performance in fulfillment costs. (d) Performance in consolidation level.

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As expected, all three methods exhibit lower average objective values because of lower fulfillment costs as the discount factor increases (see Figure 10(c)). This aligns with the intuition that higher discounts encourage more cost-efficient fulfillment strategies. The relative decrease in fulfillment costs across the methods can be attributed to their higher consolidation levels as the discount factor increases as shown in Figure 10(d). Notably, C-SAA-CLF responds most sharply, with its consolidation rate rising the most rapidly as the discount grows, which is consistent with its pronounced improvement in worst-case objective performance.

6.9. Perturbation Analysis Under Varying Simulation Environments

A critical challenge in any simulation-based evaluation is that the simulator itself is an imperfect model of the real-world data-generating process. This is particularly true in this application, where the purely observational nature of the historical data and the resulting lack of counterfactual outcomes make it impossible to construct a perfect, clairvoyant simulator. Therefore, to rigorously assess the stability of the proposed solutions, this section performs a stress test designed to probe the framework's performance under a misspecified ground truth. This analysis is conducted by introducing systematic perturbations to the simulated delivery time deviation distributions, representing moderate to adversarial divergences from the test data. Specifically, two types of perturbations are considered: random uniform noise and a delay-specific bias applied to the simulation probability distributions. Together, these tests provide critical insight into the methods' stability under real-world distributional shifts and their robustness to the inevitable simulator-reality mismatch.

6.9.1. Random Uniform Noise Perturbation. Let the original simulation distribution for each test instance be denoted by $P^{sim} = \{p_i\}_{i=1}^{C}$. For each class $i \in [C]$, a random noise term $\delta_i \sim \mathcal{U}(-\epsilon, \epsilon)$ is added to the probability p_i, resulting in $\tilde{p}_i = p_i + \delta_i$. Negative values are clipped: $\tilde{p}_i = \max(\tilde{p}_i, 0)$; the resulting vector is normalized to form a valid probability distribution: $\tilde{p}_i = \tilde{p}_i / \sum_j \tilde{p}_j$.

Figure 11 reports the average and worst-case realized objectives as the noise level ϵ increases. In the average case (Figure 11(a)), both C-RO variants degrade more gradually than C-SAA, with C-RO-B-CLF becoming the top performer beyond $\epsilon = 10$. In the worst-case scenario (Figure 11(b)), whereas all methods experience performance degrade, the ranking remains unchanged; C-RO-D achieves the lowest realized objective followed by C-RO-B and then C-SAA. These results indicate the superior robustness of the C-RO methods under uniform noise.

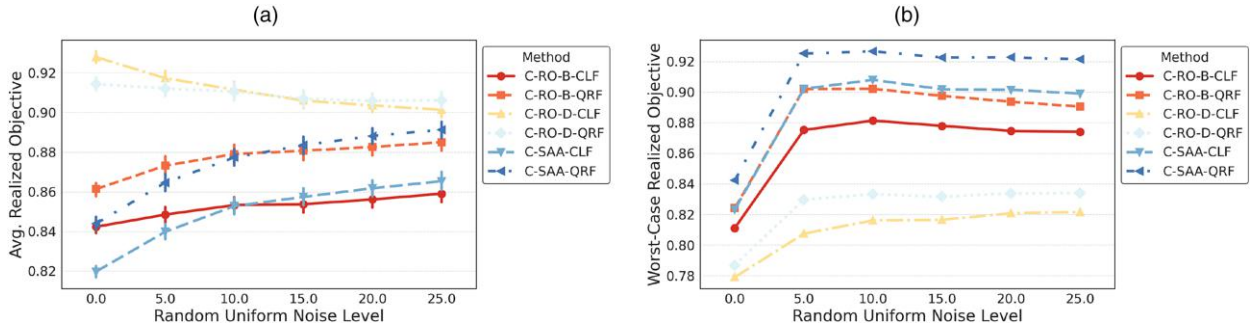
6.9.2. Delay-Specific Bias Perturbation. Define the set of late delay classes $C_{\text{late}} = \{i : \xi_i > 0\}$. A multiplicative bias η is applied to those classes: $\tilde{p}_i = p_i \cdot (1 + \eta)$. The probabilities for classes not in C_{late} remain unchanged. The resulting vector is clipped to the interval $[0, 1]$: $\tilde{p}_i = \max(\min(\tilde{p}_i, 1), 0)$ followed by normalization.

Figure 12 shows performance metrics for bias levels up to 250%. In the average case (Figure 12(a)), all CSO methods demonstrate improved performance relative to the Greedy baseline, which deteriorates rapidly. This trend is evidenced by the overall decrease in normalized objective values as the bias parameter η increases. Notably, C-RO-B overtakes C-SAA once η exceeds 150%, highlighting its robustness under adversarial delay distortions. In the worst-case objective (Figure 12(b)), C-RO-D consistently achieves the best performance followed by C-RO-B-CLF. Overall, the C-RO methods exhibit lower variability in performance compared with C-SAA, indicating enhanced stability under adversarial delay conditions.

6.10. Computational Scalability of the CSO Methods

The computational scalability of the proposed framework was evaluated by measuring both the change in computation time (model construction + solve time)

Figure 11. (Color online) Performance Across Different Random Uniform Noise Perturbations



Notes. (a) Change in average (Avg.) objective. (b) Change in worst-case objective.

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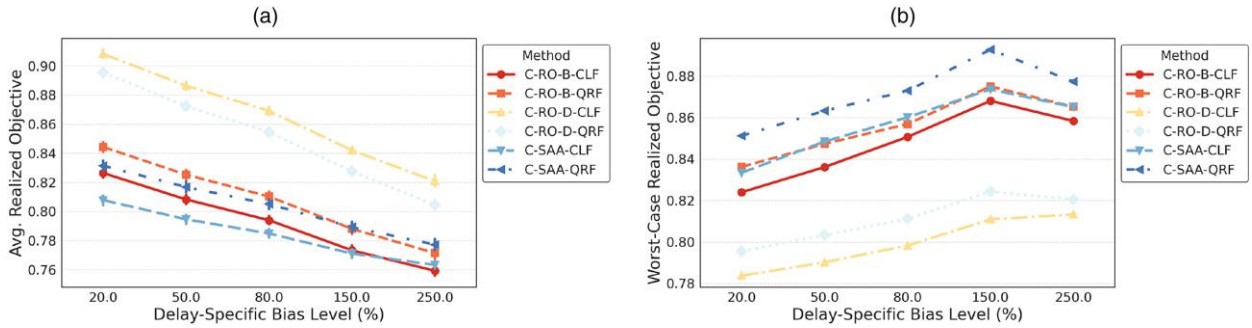
Figure 12. (Color online) Performance Across Different Delay-Specific Bias Perturbations



Notes. (a) Change in average (Avg.) objective. (b) Change in worst-case objective.

and solution quality as instance size grows. Each CSOFP instance is characterized by the number of items $|\mathcal{I}|$ and the number of location-carrier pairs (fulfillment options) $|\mathcal{K}|\times|\mathcal{L}|$. In practice, the eligibility constraint (Constraint (3)) is used to prune infeasible location-carrier pairs in advance. Then, yields the final set of eligible fulfillment options and thus, the effective problem size.

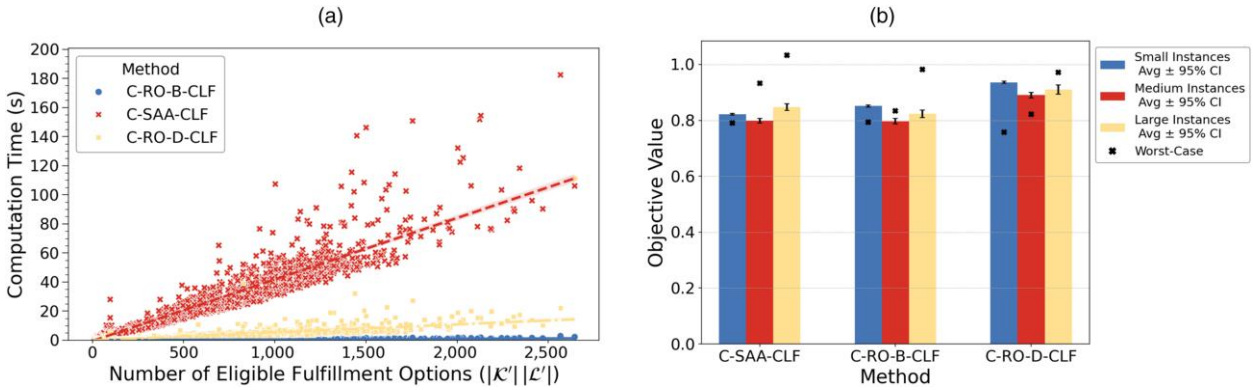
Figure 13(a) shows computation time as the number of eligible fulfillment options increases. Only the MC-CLF variants of C-RO and C-SAA are displayed because their QRF counterparts exhibit nearly identical run times. All three methods achieve optimality on every instance within minutes, confirming their practical applicability. In particular, C-RO-B solves most instances in under 1 second (never exceeding 3 seconds), and C-RO-D solves most in under 10 seconds.

These timing differences reflect each method's formulation. Compared with the nominal problem, the robust counterpart in C-RO-B introduces only continuous variables whose count grows linearly with $|\mathcal{K}'|$ and $|\mathcal{L}'|$. By contrast, C-RO-D adds N extra constraints from sampling, and C-SAA requires solving Q independent problems whose objective function complexity scales with the sample size N_1. Parallel computation of the Q subproblems can further reduce C-SAA's run time.

For completeness, a comparison of computation times against the baseline models is provided (see Online Appendix K). The Greedy and PTO methods are computationally cheaper, solving most instances in fractions of a second. However, the proposed CSO methods, particularly the highly efficient C-RO-B, remain well within practical time limits for real-world deployment (Figure 13(a)). The substantial improvements in solution quality (Table 3) justify this modest increase in computational overhead, representing a favorable trade-off between performance and speed.

Because single-item orders account for the majority of order volume, multi-item orders (what the methods are designed for) can be routed to a dedicated fulfillment optimization engine without disrupting standard operations. Under normal load, any of the three methods may be applied; under peak loads—such as holiday surges—C-RO-B is recommended to preserve fast solve times while maintaining high solution quality. Concurrent batches of orders can be processed in parallel when computational resources permit.

Figure 13. (Color online) Computational Scalability Analysis



Notes. (a) Change in the computation time vs. the number of eligible fulfillment options. (b) Objective value by instance categories. Avg stands for average; CI stands for confidence interval.

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Figure 13(b) reports average and worst-case objective values for small (≤400 eligible options), medium (≤1,000 eligible options), and large (≤3,000 eligible options) instances. Average performance remains uniformly high across all size categories, whereas worst-case deviations grow with instance size. Collectively, these results demonstrate near-linear scalability in computation time, consistent high solution quality, and robustness across problem sizes, providing empirical evidence of the framework's practical scalability.

6.11. Managerial and Operational Implications

The proposed fulfillment framework offers actionable guidance for practitioners seeking to align fulfillment strategies with demand variability, service-level requirements, and computational constraints. Depending on the operating context, different methods are preferable. When demand is stable and the goal is to minimize average fulfillment cost, C-SAA-CLF provides the most cost-effective solution. Under moderate run-time constraints, C-RO-B-CLF offers a balanced trade-off between efficiency and robustness. In contrast, during high-stakes periods such as holiday promotions or flash sales—when the cost of service-level violations is high—C-RO-D-CLF and C-RO-B-QRF (with 99% coverage) are better suited to mitigate tail risks and ensure reliability. This scenario-driven approach allows e-commerce platforms to dynamically adapt fulfillment strategies to shifting conditions while leveraging a unified optimization infrastructure. As retailers increasingly face variable demand patterns and rising customer expectations, the ability to flexibly match fulfillment strategy to context becomes a key operational advantage.

Empirical evaluations indicate that the proposed methods can reduce combined fulfillment cost and delivery-timeliness penalties by an average of up to 18% per order under the default settings. To illustrate the potential impact, consider a retailer operating at a scale comparable with the industrial partner, with per-order costs ranging from \$5 to \$30 and monthly volumes of 0.1–1 million multi-item orders. Under these scenarios, the corresponding annual savings could range from several million to tens of millions of dollars. Moreover, this framework can be readily extended to single-item orders, which could yield further savings with minimal incremental investment.

7. Conclusion

This paper introduces a generic contextual stochastic optimization framework for data-driven decision making when only observational data, hence missing counterfactual costs, are available. The framework is deliberately modular. First, any algorithm capable of returning a full predictive distribution can serve as the *contextual distribution oracle*. In the empirical study, calibrated multiclass classifiers and tree-based quantile regression forests were selected to capture the discrete, ordered nature of delivery time deviations. Second, the oracle feeds directly into two tractable optimization engines: (i) a *contextual sample average approximation* for risk-neutral objectives and (ii) a *contextual robust optimization* formulation for risk-averse objectives. Because the learning and optimization layers communicate only through the oracle, the template can be readily translated to other operations management settings.

When applied to an omnichannel multicourier order fulfillment problem, the CSO framework reduced the combined cost-service objective by up 18% on average and up to 22% in the simulated worst case on an industry-scale data set, implying annual savings from several to tens of millions of dollars. This framework allows practitioners to tune robustness—raising it during promotional peaks and relaxing it in steadier periods—while still exploiting item-consolidation discounts and heterogeneous carrier performance in the dynamic e-commerce landscape.

Although the paper delivers an end-to-end prototype with demonstrable impact, several avenues remain open.

- Performance guarantees under partial feedback. Establish theoretical performance guarantees, such as worst-case regret and postdecision surprise, for CSO methods under the partial-feedback setting. Linking causal inference techniques with SAA and RO theory can be a promising direction, thereby quantifying the value of additional data when counterfactual costs remain unobserved.
- Hybrid distributional forecasters. Develop distributional models that fuse the calibration strength of classification with the sharp tails of quantile regression, yielding tighter and more informative uncertainty sets.
- End-to-end predict-then-prescribe learning for observational data. Incorporate downstream optimization loss directly into training, extending recent "integrated predict-then-optimize" approaches (Qi et al. 2021, Elmachtoub and Grigas 2022) to settings where data are purely observational.
- Multiperiod, inventory-aware CSO. Extend the framework to jointly optimize fulfillment, shipping, and stocking over rolling horizons, capturing inventory replenishment spillovers, and elevate the model from a myopic to a long-term strategic planning tool.
- Generalization to broader problem classes. Extending the framework to broader settings, like network flow optimization, is a key direction for future work. The sampling-based C-SAA and C-RO-D approaches can be directly generalized. However, the C-RO-B formulation's tractability relies on the specific problem structure, and adapting it to more complex constraints would require new reformulations.

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